Topic 4: Solved Problems

Step-by-step solutions of problems 1-15 of the official sheet on phase diagrams (Problems T4_PhaseDiagrams I): the tie line and the lever rule of Session 7, and the invariant reactions and compounds of Session 8, written in four phases and checked against the official solutions.

DATA READ ON THE DIAGRAMS OF THE PROBLEM SHEET

System	Eutectic	α limit	β limit	Melting points
Pb-Sn (sheet p. 1)	61.9 wt% Sn, 183 °C	19.2 wt% Sn	97.5 wt% Sn	327 °C (Pb), 232 °C (Sn)
Ag-Cu (sheet p. 2)	28.1 wt% Cu, 779.1 °C	8.8 wt% Cu	92 wt% Cu	961.93 °C (Ag), 1084.87 °C (Cu)

`Problem 1` `Problems T4_PhaseDiagrams I.pdf #1` `Lever rule · two equations`

Phase Limits from the Phase Fractions of Two Alloys

`Original statement · Problems T4_PhaseDiagrams I.pdf`

1. In the phase diagram of two hypothetic metals, A and B, there is a phase α rich in A and a phase β rich in B. Given the mass fraction of two alloys for a specific temperature (table), determine the composition of the phase limits (or solid solubility limits) of the α and β phases at such temperature.

Alloy composition	*Fraction of α phase*	*Fraction of β phase*
30% B	*0.78*	*0.22*
65% B	*0.36*	*0.64*

Official Solution: $C_\alpha = 11.7\%$, $C_\beta = 95\%$

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Equilibrium:** both alloys are at the same temperature, in equilibrium, and both contain α and β only, so both lie in the α + β two-phase region [Session 7 Slide 26].
2. **One tie line:** at a fixed temperature and pressure, a two-component, two-phase system has $F = C - P + 1 = 2 - 2 + 1 = 1$ degree of freedom, already used by the temperature [Session 7 Slide 13]. The compositions of α and β are therefore the same for both alloys: C_α and C_β are the two ends of one tie line.
3. **Mass fractions** obey the lever rule [Session 7 Slide 16].

Data

Quantity	Alloy 1	Alloy 2
Overall composition C_0 (% B)	30	65
W_α	0.78	0.36
W_β	0.22	0.64

Unknowns: C_α and C_β (% B). Two unknowns and two independent equations (one per alloy; $W_\beta = 1 - W_\alpha$ adds nothing new).

PHASE 2 · FRAMES & STRATEGY

The only frame is the phase diagram itself: composition in % B along the horizontal axis and temperature along the vertical axis. All the work is done on one horizontal line (the tie line at the given temperature), so it is one-dimensional and no change of basis is involved.

Why the lever rule. It is the tool that relates the amounts of the phases to the positions of the alloy and of the phase limits on the tie line [Session 7 Slide 16]:

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$$

Writing it once for each alloy gives two linear equations in C_α and C_β.

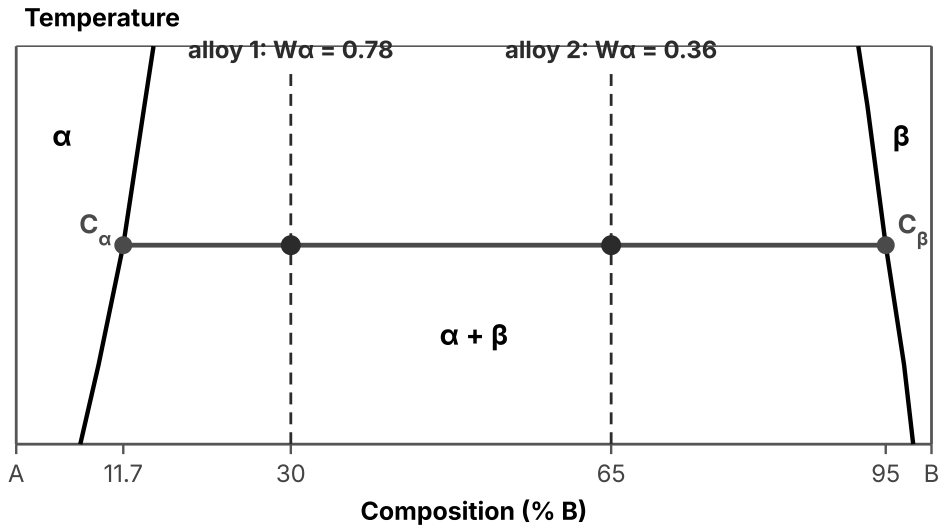


Figure 1. Both alloys are in the α + β region at the same temperature, so they share the tie line from C_α to C_β; the two lever-rule equations give C_α = 11.7 % B and C_β = 95 % B.

PHASE 3 · STEP-BY-STEP DERIVATION

1. The two equations Call $D = C_\beta - C_\alpha$ the length of the tie line. Multiplying the lever rule by D:

$$W_\alpha D = C_\beta - C_0$$

Alloy 1 ($C_0 = 30$, $W_\alpha = 0.78$) and alloy 2 ($C_0 = 65$, $W_\alpha = 0.36$):

$$0.78\,D = C_\beta - 30 \qquad (1)$$

$$0.36\,D = C_\beta - 65 \qquad (2)$$

2. Length of the tie line Subtracting (2) from (1) eliminates C_β:

$$0.78\,D - 0.36\,D = (C_\beta - 30) - (C_\beta - 65) \;\Rightarrow\; 0.42\,D = 35 \;\Rightarrow\; D = \frac{35}{0.42} = 83.33\ \%\mathrm{B}$$

3. Limit of β From (1):

$$C_\beta = 30 + 0.78 \times 83.33 = 30 + 65.00 = 95.0\ \%\mathrm{B}$$

4. Limit of α

$$C_\alpha = C_\beta - D = 95.0 - 83.33 = 11.67 \approx 11.7\ \%\mathrm{B}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **β fractions** (not used in the solution): $W_\beta = \dfrac{C_0 - C_\alpha}{C_\beta - C_\alpha}$ gives $\dfrac{30 - 11.67}{83.33} = 0.22$ and $\dfrac{65 - 11.67}{83.33} = 0.64$, exactly the values of the table.
- **Mass balance of B** for alloy 1: $0.78 \times 11.67 + 0.22 \times 95.0 = 9.10 + 20.90 = 30.0\ \%\text{B} = C_0$.
- **Order:** $C_\alpha = 11.7 < 30 < 65 < 95 = C_\beta$, so both alloys are indeed inside the α + β region, as assumed.
- **Units:** all compositions are in % B (mass), fractions are dimensionless; each equation is homogeneous in % B.

Interpretation At this temperature α can dissolve at most 11.7 % B and β at most 5 % A (95 % B). The alloy closer to the α limit (30 % B) has more α, as the lever rule predicts.

Check against the official solution

The official solution gives $C_\alpha = 11.7\%$ and $C_\beta = 95\%$; the result agrees. The official solutions PDF has no worked page for this problem; only the key of the statement is available.

`Problem 2` `Problems T4_PhaseDiagrams I.pdf #2` `Pb-Sn · hypereutectic`

Primary and Total β in a Pb-Sn Alloy

`Original statement · Problems T4_PhaseDiagrams I.pdf`

2. From 2.8 kg of a Sn-Pb alloy, is it possible to have 2.21 kg of primary β and 2.53 kg of β total at (183 – ΔT) °C? Justify your answer.

`Official Solution: Yes, for an alloy with composition` $C_0 = 90\%$ `B`

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Equilibrium cooling** with the Pb-Sn diagram of the problem sheet (Problems T4_PhaseDiagrams I.pdf, p. 1): eutectic at $C_e = 61.9$ wt% Sn and 183 °C; at 183 °C the α limit is $C_\alpha = 19.2$ wt% Sn and the β limit $C_\beta = 97.5$ wt% Sn.
2. **Primary β** exists only in a **hypereutectic** alloy ($61.9 < C_0 < 97.5$ wt% Sn): it is the β that forms in the β + L region before the eutectic reaction [Session 7 Slide 31].
3. Just below the eutectic temperature the primary β keeps the amount it had just above it, and the liquid present just above becomes the eutectic [Session 7 Slide 32].
4. "% B" in the key means wt% Sn (B is the Sn-rich side of the diagram).

Data

Quantity	Value
Mass of alloy m	2.8 kg
Mass of primary β $m_{\beta'}$	2.21 kg
Mass of total β m_β (at 183 – ΔT)	2.53 kg
C_α, C_e, C_β at 183 °C	19.2, 61.9, 97.5 wt% Sn

Unknown: the composition C_0. There is one unknown and two conditions, so the question is whether **both** masses can be met by the same C_0.

PHASE 2 · FRAMES & STRATEGY

The frame is the Pb-Sn diagram, composition in wt% Sn and temperature in °C; the work is on the horizontal tie lines just above and just below 183 °C, so it is one-dimensional and no change of basis is needed.

Strategy. The primary β fixes C_0 through the lever rule on the short tie line between the eutectic liquid and β (microconstituents). The total β is then **predicted** with the lever rule on the long tie line between α and β (phases), and compared with 2.53 kg. If they agree, the answer is yes.

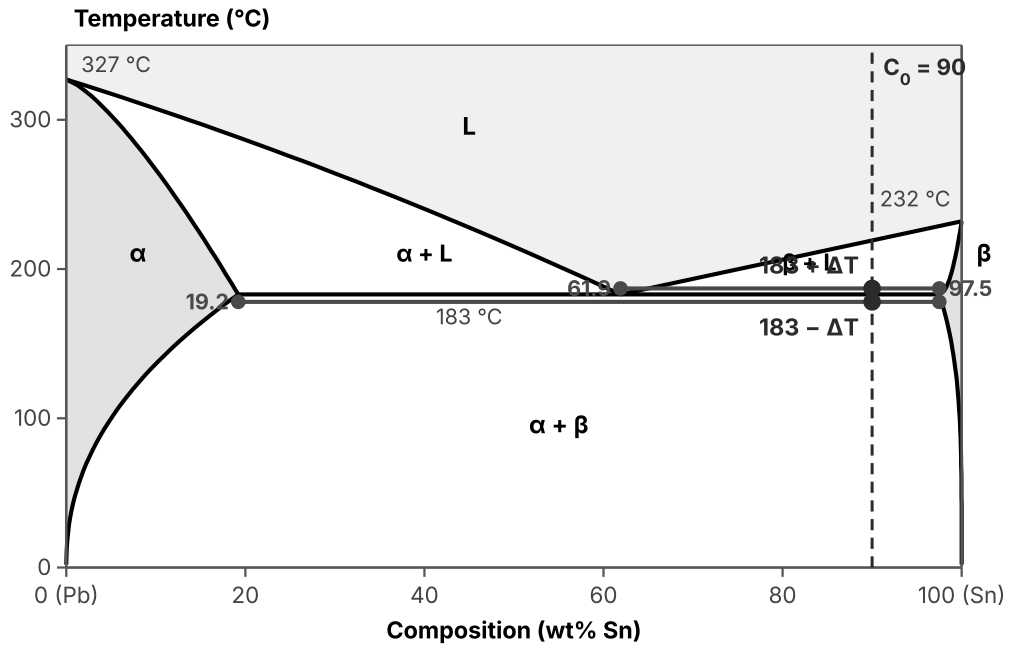


Figure 2. Pb-Sn diagram of the problem sheet (eutectic 61.9 wt% Sn at 183 °C, α 19.2 and β 97.5 wt% Sn). The alloy with 79 % primary β is the hypereutectic Pb-90 wt% Sn; the tie lines just above and below 183 °C give the primary β and the total β.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Mass fraction of primary β

$$W_{\beta'} = \frac{m_{\beta'}}{m} = \frac{2.21\ \text{kg}}{2.8\ \text{kg}} = 0.7893$$

2. Composition of the alloy Just above 183 °C the alloy is β (97.5 wt% Sn) plus liquid of eutectic composition (61.9 wt% Sn). The lever rule gives the β fraction, which is the primary β [Session 7 Slide 32]:

$$W_{\beta'} = \frac{C_0 - C_e}{C_\beta - C_e} = \frac{C_0 - 61.9}{97.5 - 61.9} = \frac{C_0 - 61.9}{35.6}$$

Solving for C_0:

$$C_0 = 61.9 + 0.7893 \times 35.6 = 61.9 + 28.10 = 90.0 \text{ wt\% Sn}$$

$61.9 < 90.0 < 97.5$: the alloy is hypereutectic, as primary β requires.

3. Total β predicted for this alloy Just below 183 °C the phases are α (19.2 wt% Sn) and β (97.5 wt% Sn):

$$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{90.0 - 19.2}{97.5 - 19.2} = \frac{70.8}{78.3} = 0.9042$$

$$m_\beta = W_\beta \, m = 0.9042 \times 2.8 \text{ kg} = 2.532 \text{ kg} \approx 2.53 \text{ kg}$$

4. Answer Both conditions are met by the same alloy: **yes**, it is possible, with a Pb-90 wt% Sn alloy.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Eutectic part:** mass of eutectic $= 2.8 - 2.21 = 0.59$ kg; β inside it $= 2.53 - 2.21 = 0.32$ kg, a fraction $0.32/0.59 = 0.54$. The eutectic of the sheet diagram has $W_{\beta,e} = (61.9 - 19.2)/(97.5 - 19.2) = 42.7/78.3 = 0.545$ of β, and $0.545 \times 0.59 = 0.32$ kg: consistent.
- **Units:** masses in kg, fractions dimensionless, compositions in wt% Sn.

Interpretation The two masses are not independent: once the primary β fixes the composition, the total β is fixed too. A different total β (say 2.40 kg) would have been impossible for any alloy.

Check against the official solution

The official key ("Yes, for an alloy with composition $C_0 = 90\%$") agrees. The worked page (Solution problems Phase diagrams I.pdf, p. 6) finds $C_0 = 90\%$ from the primary β only; the check of the total β (2.53 kg) above completes the justification. The page writes the fraction as "079"; it is 0.79.

Problem 3 | Problems T4_PhaseDiagrams I.pdf #3 | Pb-Sn · hypereutectic

Phases and Microconstituents of a Pb-80 wt% Sn Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf

3. Using the Pb-Sn phase diagram for an alloy composition of Pb-80%Sn, indicate at the temperatures of $T_1 = 250$ ºC, $T_2 = 183 + \Delta T$ ºC, $T_3 = 183 - \Delta T$ ºC, $T_4 = 100$ ºC: a) The phases present, chemical composition of the phases and mass fraction. b) The micro constituents, chemical composition and mass fractions. Note the agreement that exists between the data obtained in (a) and (b).

Official Solution: a) T_1: $C_L = 80\%$ **Sn**, $w_L = 1$; T_2: $C_L = 61.9\%$ **Sn**, $C_\beta = 97.5\%$ **Sn**, $w_L = 0.49$; T_3: $C_\alpha = 19.2\%$ **Sn**, $C_\beta = 97.5\%$ **Sn**, $w_\alpha = 0.22$; T_4: $C_\alpha = 5\%$ **Sn**, $C_\beta = 100\%$ **Sn**, $w_\alpha = 0.21$; **b)** $C_{\beta'} = 97.5\%$ **Sn**, $C_E = 61.9\%$ **Sn**, $w_E = 0.49$, $w_{\beta'} = 0.51$

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling**, Pb-Sn diagram of the problem sheet (p. 1): eutectic at 61.9 wt% Sn and 183 °C; at 183 °C, α of 19.2 and β of 97.5 wt% Sn.
2. Values at 250 °C and 100 °C are **readings of the figure**: at 250 °C the alloy is above the liquidus on the Sn side (which goes from 183 °C at 61.9 wt% Sn to 232 °C at 100 wt% Sn); at 100 °C the solvus lines give α of about 5 and β of about 100 wt% Sn.
3. $C_0 = 80$ wt% Sn lies between 61.9 and 97.5: a **hypereutectic** alloy, so the primary phase is β [Session 7 Slide 31].

Data

Temperature	Region	Phase compositions (wt% Sn)
$T_1 = 250$ °C	L	$C_L = 80$
$T_2 = 183 + \Delta T$	β + L	$C_L = 61.9$, $C_\beta = 97.5$
$T_3 = 183 - \Delta T$	α + β	$C_\alpha = 19.2$, $C_\beta = 97.5$
$T_4 = 100$ °C	α + β	$C_\alpha = 5$, $C_\beta = 100$

PHASE 2 · FRAMES & STRATEGY

The frame is the Pb-Sn diagram (wt% Sn, °C). At each temperature the work is on one horizontal tie line, so it is one-dimensional; no change of basis is needed.

Method. (a) Tie line for the compositions, lever rule for the fractions [Session 7 Slides 15-16]. (b) Below the eutectic temperature, the lever rule on the short tie line (β to eutectic liquid) gives the microconstituents, and the long tie line (α to β) gives the phases [Session 7 Slide 32]; comparing both answers part (b)'s "agreement".

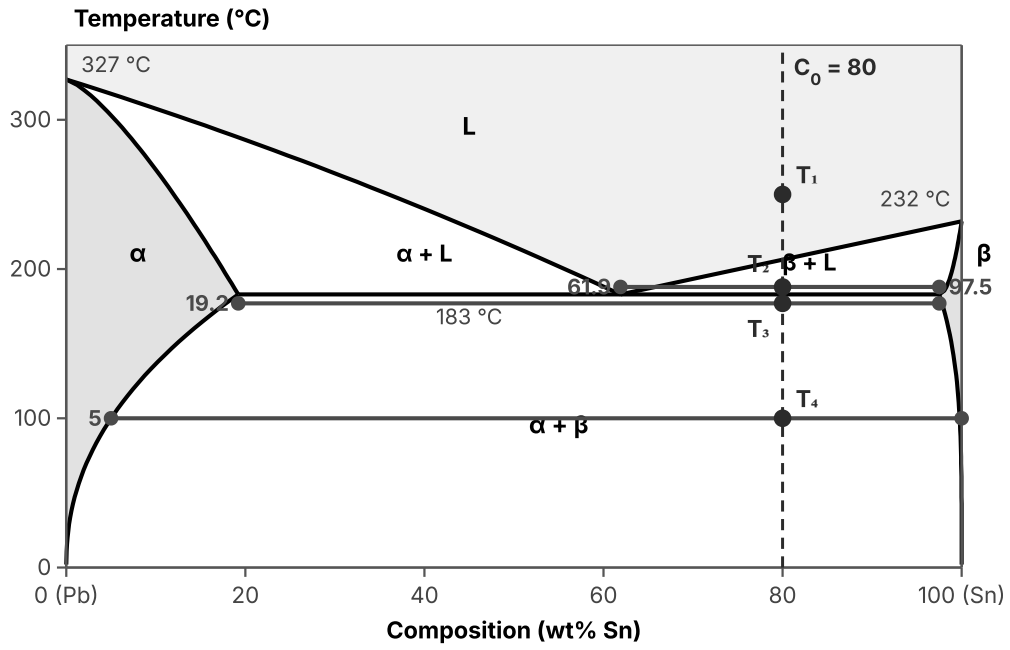


Figure 3. Pb-80 wt% Sn at T_1 = 250 °C (liquid), T_2 = 183 + ΔT (β + L), T_3 = 183 − ΔT (α + β) and T_4 = 100 °C (α + β, with α of 5 and β of 100 wt% Sn, a reading of the figure). Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 250$ °C. Above the liquidus: one phase, liquid with the composition of the alloy. $C_L = 80$ wt% Sn, $W_L = 1$.

$T_2 = 183 + \Delta T$**.** Region β + L; tie line from $C_L = 61.9$ to $C_\beta = 97.5$ (length $97.5 - 61.9 = 35.6$):

$$W_\beta = \frac{C_0 - C_L}{C_\beta - C_L} = \frac{80 - 61.9}{97.5 - 61.9} = \frac{18.1}{35.6} = 0.508 \qquad W_L = \frac{C_\beta - C_0}{C_\beta - C_L} = \frac{97.5 - 80}{35.6} = \frac{17.5}{35.6} = 0.492$$

$T_3 = 183 - \Delta T$**.** Region α + β; tie line from 19.2 to 97.5 (length 78.3):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{97.5 - 80}{97.5 - 19.2} = \frac{17.5}{78.3} = 0.224 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{80 - 19.2}{78.3} = \frac{60.8}{78.3} = 0.776$$

$T_4 = 100$ °**C.** Region α + β; tie line from 5 to 100 (length 95):

$$W_\alpha = \frac{100 - 80}{100 - 5} = \frac{20}{95} = 0.211 \qquad W_\beta = \frac{80 - 5}{95} = \frac{75}{95} = 0.789$$

b) Microconstituents At T_1 and T_2 the microconstituents are the phases themselves (liquid; primary β + liquid).

At T_3 there are two microconstituents: **primary β** (97.5 wt% Sn), formed in the β + L region, and **eutectic** (overall 61.9 wt% Sn), formed from the liquid present at T_2. With $P = C_0 - C_e$ and $Q = C_\beta - C_0$ on the short tie line:

$$W_{\beta'} = \frac{C_0 - C_e}{C_\beta - C_e} = \frac{80 - 61.9}{97.5 - 61.9} = 0.508 = W_\beta(T_2) \qquad W_E = \frac{C_\beta - C_0}{C_\beta - C_e} = \frac{17.5}{35.6} = 0.492 = W_L(T_2)$$

Phases inside the eutectic:

$$W_{\beta,E} = W_\beta(T_3) - W_{\beta'} = 0.776 - 0.508 = 0.268 \qquad W_{\alpha,E} = W_\alpha(T_3) = 0.224$$

(all the α is in the eutectic, since the primary phase is β).

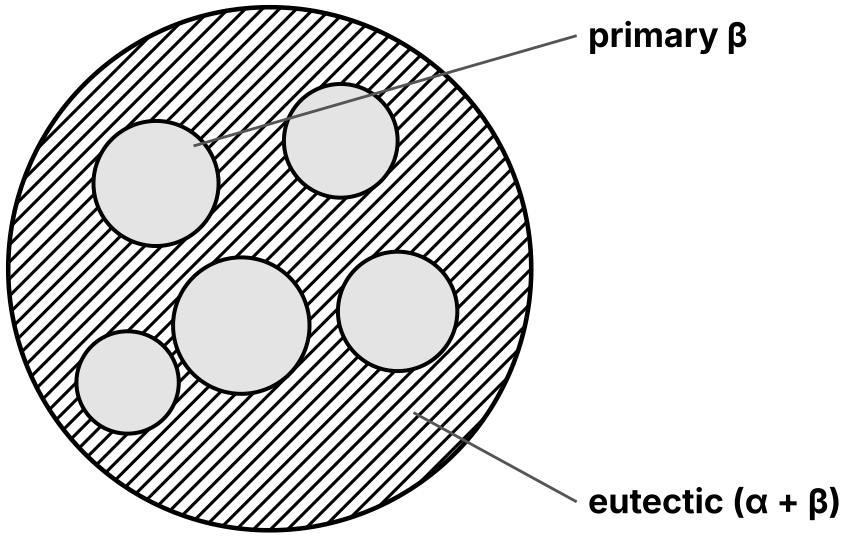


Figure 4. Pb-80 wt% Sn below the eutectic temperature: primary β (51 %) in the eutectic α + β (49 %).

PHASE 4 · CHECKS & INTERPRETATION

Agreement between (a) and (b)

- $W_{\beta,E} + W_{\alpha,E} = 0.268 + 0.224 = 0.492 = W_E$: the phases of the eutectic add up to the eutectic.
- Inside the eutectic, α is $0.224/0.492 = 0.455$, the same as the eutectic product of the sheet diagram, $(97.5 - 61.9)/(97.5 - 19.2) = 35.6/78.3 = 0.455$.
- Mass balance at T_3: $0.224 \times 19.2 + 0.776 \times 97.5 = 4.30 + 75.66 = 80.0$ wt% Sn.

Interpretation Below 183 °C the alloy is about half primary β and half eutectic. On further cooling to 100 °C the solvus lines move outwards (α purer in Pb, β practically pure Sn) and the amounts of the phases change slightly (α from 0.224 to 0.211) by solid-state precipitation, while the microconstituents keep their shape.

Check against the official solution

All the values agree with the key ($w_L = 0.49$, $w_\alpha = 0.22$, $w_\alpha = 0.21$; $w_E = 0.49$, $w_{\beta'} = 0.51$).

ERRATUM IN THE WORKED SOLUTION

Solution problems Phase diagrams I.pdf, p. 7, writes "T2; phases α and L"; the phases at T_2 are **β and L** (the alloy is hypereutectic). On the same page the T_3 formula is printed with the denominator $97.5 - 61.9$, but the value 0.22 corresponds to $97.5 - 19.2 = 78.3$: $17.5/35.6$ would give 0.49, while $17.5/78.3 = 0.22$. The numbers of the page are right; the labels and that denominator are not.

Problem 4 | Problems T4_PhaseDiagrams I.pdf #4 | Pb-Sn · hypoeutectic

Phases and Microconstituents of a Pb-30 wt% Sn Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf

4. Using the Pb-Sn phase diagram for an alloy composition of Pb-30%Sn, indicate at the temperatures of $T_1 = 250$ ºC, $T_2 = 183 + \Delta T$ ºC, $T_3 = 183 - \Delta T$ ºC, $T_4 = 100$ ºC: a) The phases present, chemical composition of the phases and mass fraction. b) The micro constituents, chemical composition and mass fractions. Note the agreement that exists between the data obtained in (a) and (b).

Official Solution: a) T_1: $C_L = 36\%$ Sn, $C_\alpha = 12\%$ Sn, $w_L = 0.75$; T_2: $C_L = 61.9\%$ Sn, $C_\alpha = 19.2\%$ Sn, $w_L = 0.25$; T_3: $C_\alpha = 19.2\%$ Sn, $C_\beta = 97.5\%$ Sn, $w_\alpha = 0.86$; T_4: $C_\alpha = 5\%$ Sn, $C_\beta = 100\%$ Sn, $w_\alpha = 0.74$; **b)** T_3: $C_{\alpha'} = 19.2\%$ Sn, $C_E = 61.9\%$ Sn, $w_E = 0.25$, $w_{\alpha'} = 0.75$

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling**, Pb-Sn diagram of the problem sheet (p. 1): eutectic at 61.9 wt% Sn and 183 °C; α of 19.2 and β of 97.5 wt% Sn at 183 °C.
2. **Readings of the figure:** at 250 °C the tie line meets the α solidus at 12 wt% Sn and the liquidus at 36 wt% Sn; at 100 °C the solvus lines give α of 5 and β of 100 wt% Sn (the same readings as the official solution).
3. $19.2 < C_0 = 30 < 61.9$: a **hypoeutectic** alloy, so the primary phase is α [Session 7 Slide 30]. This is the same kind of alloy as the hypoeutectic example of the slides.

Data

Temperature	Region	Phase compositions (wt% Sn)
$T_1 = 250$ °C	α + L	$C_\alpha = 12$, $C_L = 36$
$T_2 = 183 + \Delta T$	α + L	$C_\alpha = 19.2$, $C_L = 61.9$
$T_3 = 183 - \Delta T$	α + β	$C_\alpha = 19.2$, $C_\beta = 97.5$
$T_4 = 100$ °C	α + β	$C_\alpha = 5$, $C_\beta = 100$

PHASE 2 · FRAMES & STRATEGY

Frame: the Pb-Sn diagram (wt% Sn, °C); one horizontal tie line per temperature, so no change of basis is needed.

Method. Tie line and lever rule at each temperature [Session 7 Slides 15-16]; below 183 °C, the short tie line (α to eutectic liquid) gives the microconstituents and the long one (α to β) the phases [Session 7 Slide 32].

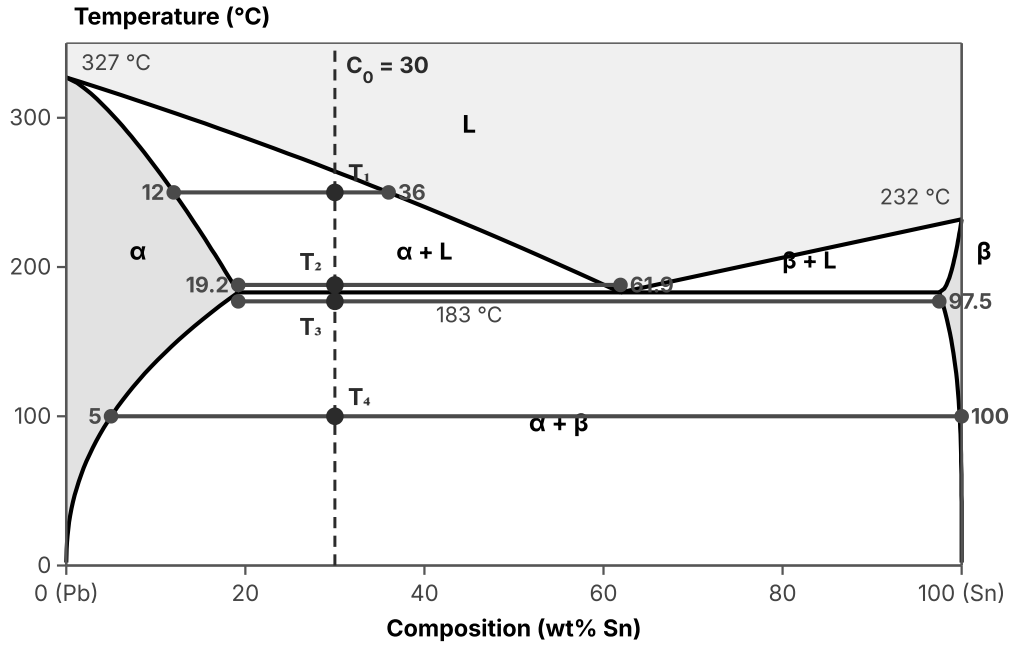


Figure 5. Pb-30 wt% Sn at T_1 = 250 °C (α of 12 and L of 36 wt% Sn, a reading of the figure), T_2 = 183 + ΔT (α + L), T_3 = 183 − ΔT (α + β) and T_4 = 100 °C. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 250$ °C. α + L; tie line from 12 to 36 (length 24):

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{30 - 12}{36 - 12} = \frac{18}{24} = 0.750 \qquad W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{36 - 30}{24} = \frac{6}{24} = 0.250$$

$T_2 = 183 + \Delta T$**.** α + L; tie line from 19.2 to 61.9 (length 42.7):

$$W_L = \frac{30 - 19.2}{61.9 - 19.2} = \frac{10.8}{42.7} = 0.253 \qquad W_\alpha = \frac{61.9 - 30}{42.7} = \frac{31.9}{42.7} = 0.747$$

$T_3 = 183 - \Delta T$**.** α + β; tie line from 19.2 to 97.5 (length 78.3):

$$W_\alpha = \frac{97.5 - 30}{97.5 - 19.2} = \frac{67.5}{78.3} = 0.862 \qquad W_\beta = \frac{30 - 19.2}{78.3} = \frac{10.8}{78.3} = 0.138$$

$T_4 = 100$ **°C.** α + β; tie line from 5 to 100 (length 95):

$$W_\alpha = \frac{100 - 30}{100 - 5} = \frac{70}{95} = 0.737 \qquad W_\beta = \frac{30 - 5}{95} = \frac{25}{95} = 0.263$$

b) Microconstituents At T_1 and T_2 they coincide with the phases (α + liquid). At T_3: **primary α** (19.2 wt% Sn) and **eutectic** (61.9 wt% Sn overall):

$$W_{\alpha'} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{61.9 - 30}{61.9 - 19.2} = 0.747 = W_\alpha(T_2) \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{30 - 19.2}{42.7} = 0.253 = W_L(T_2)$$

Phases inside the eutectic:

$$W_{\alpha,E} = W_\alpha(T_3) - W_{\alpha'} = 0.862 - 0.747 = 0.115 \qquad W_{\beta,E} = W_\beta(T_3) = 0.138$$

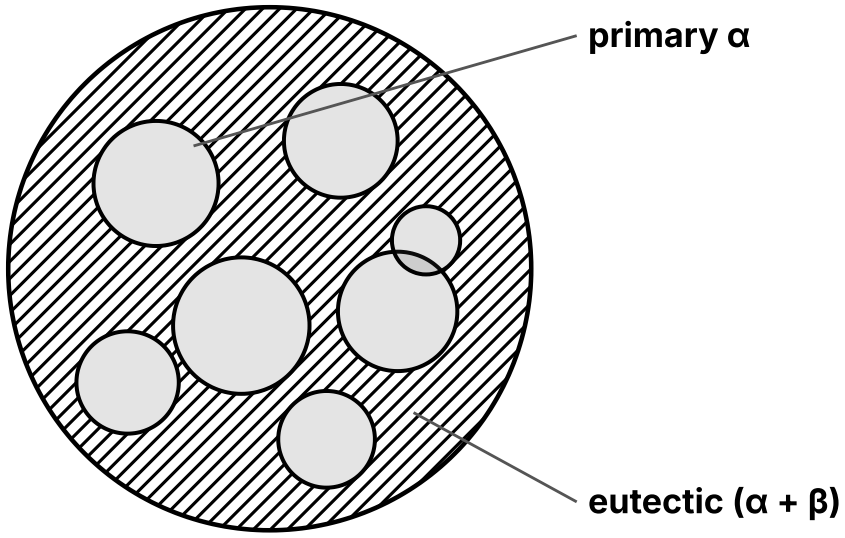


Figure 6. Pb-30 wt% Sn below the eutectic temperature: primary α (75 %) in the eutectic α + β (25 %).

PHASE 4 · CHECKS & INTERPRETATION

Agreement between (a) and (b)

- $W_{\alpha,E} + W_{\beta,E} = 0.115 + 0.138 = 0.253 = W_E$.
- Inside the eutectic, α is $0.115/0.253 = 0.455$, the eutectic product of the sheet diagram ($35.6/78.3 = 0.455$), the same as in Problem 3: the eutectic has the same make-up whatever the alloy.
- Mass balance at T_3: $0.862 \times 19.2 + 0.138 \times 97.5 = 16.55 + 13.45 = 30.0$ wt% Sn.

Interpretation The alloy is three quarters primary α in one quarter of eutectic, the structure of the hypoeutectic micrograph of Slide 30. From 250 °C to 183 °C the amount of α grows from 25 % to 75 % while the liquid gets richer in Sn, from 36 to 61.9 wt%.

Check against the official solution

The values agree with the key ($w_L = 0.75, 0.25$; $w_\alpha = 0.86, 0.74$; $w_E = 0.25$, $w_{\alpha'} = 0.75$). The α inside the eutectic is $0.862 - 0.747 = 0.115$; the worked page writes 0.11 because it subtracts the rounded values $0.86 - 0.75$.

ERRATUM IN THE WORKED SOLUTION

Solution problems Phase diagrams I.pdf, p. 10, gives for T_4 the numbers of Problem 3: $\chi_\alpha = (100 - 80)/(100 - 5) = 0.21$ and $\chi_{\beta,\text{total}} = 0.79$. For this alloy ($C_0 = 30$) the correct values are $W_\alpha = 70/95 = 0.74$ and $W_\beta = 0.26$, as the key of the statement says. The T_1 formula on p. 9 is also printed with the wrong labels ($C_o - C_\beta$ over $C_E - C_\alpha$), while its numbers, $(30 - 12)/(36 - 12) = 0.75$, are right.

Problem 5 | Problems T4_PhaseDiagrams I.pdf #5 | Ag-Cu · eutectic

The Eutectic Ag-28.1 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

5. Given an alloy Ag-28.1%Cu, indicate for the given temperatures, $T_1 = 900$ °C; $T_2 = (780 + \Delta T)$ °C; $T_3 = (780 - \Delta T)$ °C: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at T_3.

Official Solution: a) T_1: Liquid (28.1%Cu); T_2: Liquid (28.1%Cu); T_3: α(C_α = 8.8%Cu) + β(C_β = 92%Cu); $x_\alpha = 0.77$; $x_\beta = 0.23$. b) Eutectic microconstituent (28.1%Cu); $C_{\alpha E}$ = 8.8%Cu; $C_{\beta E}$ = 92%Cu

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling** with the Ag-Cu diagram of the problem sheet (p. 2): eutectic at 28.1 wt% Cu and 779.1 °C (the statement rounds it to 780 °C); at that temperature (Ag) = α contains 8.8 wt% Cu and (Cu) = β contains 92 wt% Cu. Melting points: 961.93 °C (Ag) and 1084.87 °C (Cu).
2. $C_0 = 28.1$ wt% Cu is exactly the **eutectic composition** [Session 7 Slides 26-27].

Data

Quantity	Value
C_0	28.1 wt% Cu
C_α, C_e, C_β at the eutectic temperature	8.8, 28.1, 92 wt% Cu
Eutectic temperature	779.1 °C ≈ 780 °C

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one horizontal tie line per temperature; no change of basis is needed.

Why no lever rule above 780 °C. The eutectic liquid meets both liquidus lines at their lowest point, so it never enters a two-phase L + solid region: it stays liquid down to the eutectic temperature and then transforms at once, $L \rightarrow \alpha + \beta$ [Session 7 Slide 27].

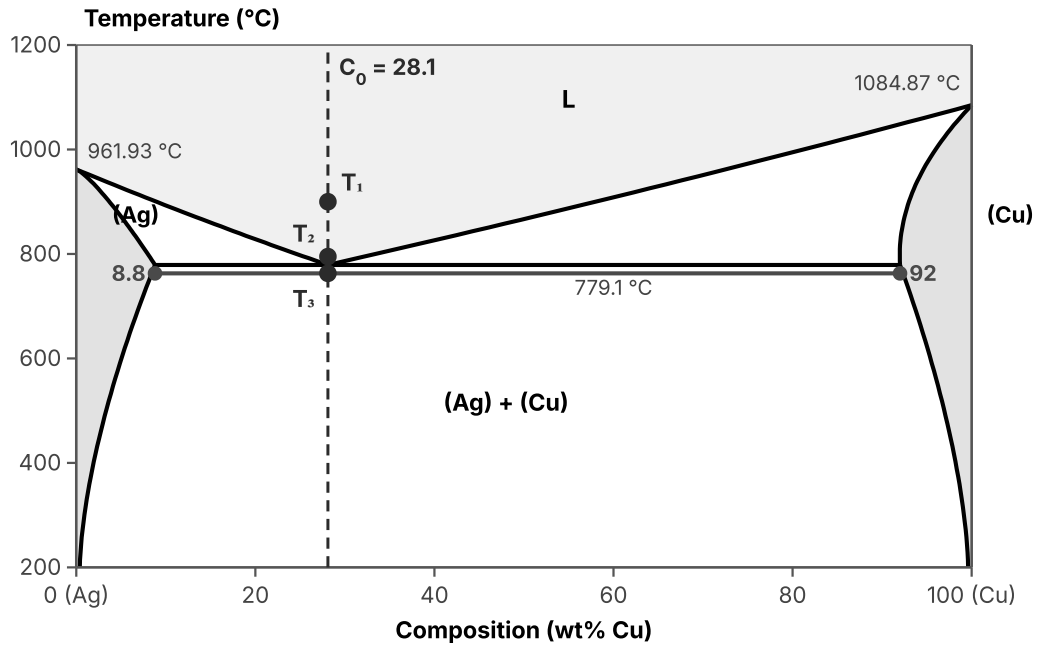


Figure 7. Ag-Cu diagram of the problem sheet. The eutectic alloy is liquid down to 779.1 °C and then becomes (Ag) of 8.8 and (Cu) of 92 wt% Cu.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 900$ °**C.** Above the liquidus: liquid, $C_L = 28.1$ wt% Cu, $W_L = 1$.

$T_2 = 780 + \Delta T$**.** Still above the eutectic point: liquid, $C_L = 28.1$ wt% Cu, $W_L = 1$.

$T_3 = 780 - \Delta T$**.** α + β; tie line from 8.8 to 92 (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{92 - 28.1}{92 - 8.8} = \frac{63.9}{83.2} = 0.768 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{28.1 - 8.8}{83.2} = \frac{19.3}{83.2} = 0.232$$

b) Microconstituents at T_3 All the liquid of T_2 became eutectic: **one microconstituent, the eutectic**, $W_E = 1$, with overall composition 28.1 wt% Cu. Inside it: α (8.8 wt% Cu) with mass fraction 0.768 and β (92 wt% Cu) with mass fraction 0.232.

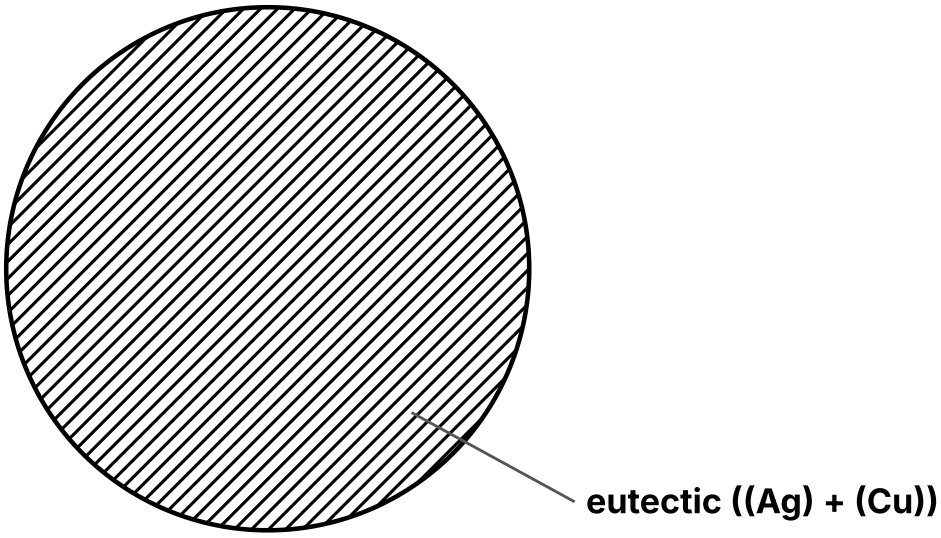


Figure 8. Ag-28.1 wt% Cu below 779 °C: 100 % eutectic, (Ag) and (Cu) in alternating layers.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- $W_\alpha + W_\beta = 0.768 + 0.232 = 1$.
- Mass balance: $0.768 \times 8.8 + 0.232 \times 92 = 6.76 + 21.34 = 28.1$ wt% Cu $= C_0$.
- Phase rule at the eutectic point (1 atm): $F = 2 - 3 + 1 = 0$, so the transformation happens at one temperature, with a plateau in the cooling curve, like a pure metal [Session 7 Slide 27].

Interpretation The eutectic alloy has the lowest melting point of the system (779 °C, well below 961.93 °C for Ag and 1084.87 °C for Cu) and solidifies at a single temperature into a fine layered mixture of a Ag-rich and a Cu-rich solid.

Check against the official solution

Agrees with the key: $x_\alpha = 0.77$, $x_\beta = 0.23$; 100 % eutectic. The official solutions PDF has no worked page for this problem.

Problem 6 | Problems T4_PhaseDiagrams I.pdf #6 | Ag-Cu · hypereutectic

The Hypereutectic Ag-80 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

6. Using the Cu-Ag phase diagram, for an alloy with composition C_0 = 80% Cu, for temperatures $T_1 = 900$ °C; $T_2 = (780 + \Delta T)$ °C; $T_3 = (780 - \Delta T)$ °C, determine: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at T_3.

Official Solution: a) T_1**:** C_L **= 58%Cu;** C_β **= 93%Cu;** $X_L = 0.37$**;** $X_\beta = 0.63$**;** T_2**:** C_L **= 28.1%Cu;** C_β **= 92%Cu;** $X_L = 0.19$**;** $X_\beta = 0.81$**;** T_3**:** C_α **= 8.8%Cu;** C_β **= 92%Cu;** $X_\alpha = 0.14$**;** $X_\beta = 0.86$**. b) eutectic microconstituent (**$X_E = 0.19$**) and proeutectic β (**$X_{\beta p} = 0.81$**)**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling**, Ag-Cu diagram of the problem sheet (p. 2): eutectic 28.1 wt% Cu at 779.1 °C ≈ 780 °C; α = (Ag) of 8.8 and β = (Cu) of 92 wt% Cu at that temperature.
2. **Reading of the figure at 900 °C:** the tie line meets the Cu-side liquidus at 58 wt% Cu and the (Cu) solidus at 93 wt% Cu (the readings of the official solution).
3. $28.1 < C_0 = 80 < 92$: a **hypereutectic** alloy, primary phase β [Session 7 Slide 31].

Data

Temperature	Region	Phase compositions (wt% Cu)
$T_1 = 900$ °C	L + β	$C_L = 58$, $C_\beta = 93$
$T_2 = 780 + \Delta T$	L + β	$C_L = 28.1$, $C_\beta = 92$
$T_3 = 780 - \Delta T$	α + β	$C_\alpha = 8.8$, $C_\beta = 92$

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one tie line per temperature; no change of basis. Lever rule for phases [Session 7 Slide 16]; short tie line (eutectic liquid to β) for the microconstituents [Session 7 Slide 32].

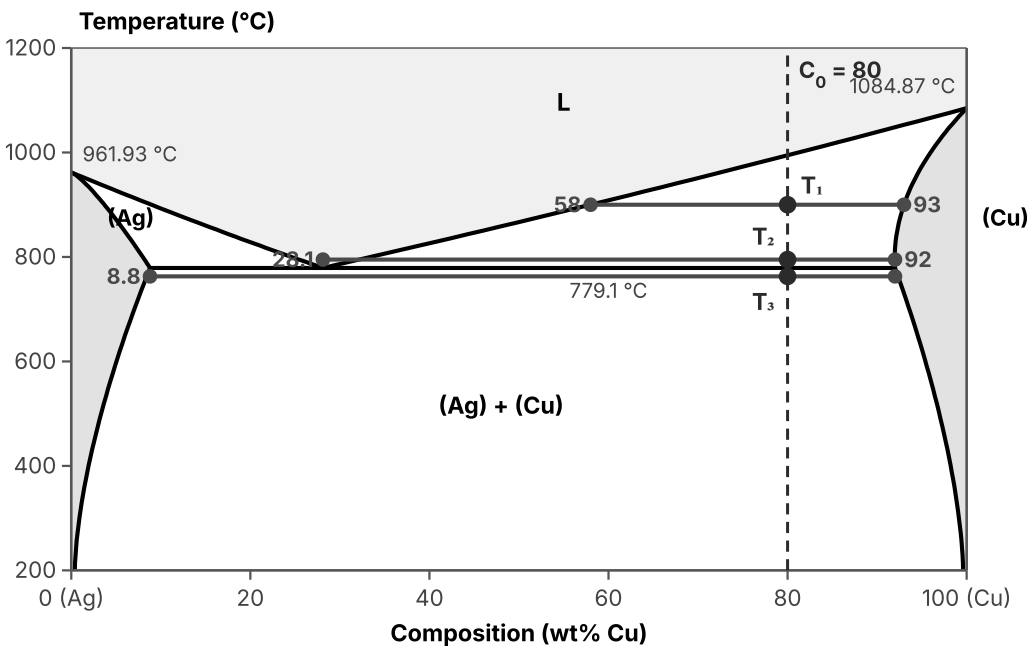


Figure 9. Ag-80 wt% Cu at T_1 = 900 °C (L of 58 and (Cu) of 93 wt% Cu, a reading of the figure), T_2 = 780 + ΔT and T_3 = 780 − ΔT. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 900$ **°C** (tie line length $93 - 58 = 35$):

$$W_L = \frac{C_\beta - C_0}{C_\beta - C_L} = \frac{93 - 80}{93 - 58} = \frac{13}{35} = 0.371 \qquad W_\beta = \frac{C_0 - C_L}{C_\beta - C_L} = \frac{80 - 58}{35} = \frac{22}{35} = 0.629$$

$T_2 = 780 + \Delta T$ (length $92 - 28.1 = 63.9$):

$$W_L = \frac{92 - 80}{92 - 28.1} = \frac{12}{63.9} = 0.188 \qquad W_\beta = \frac{80 - 28.1}{63.9} = \frac{51.9}{63.9} = 0.812$$

$T_3 = 780 - \Delta T$ (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{92 - 80}{92 - 8.8} = \frac{12}{83.2} = 0.144 \qquad W_\beta = \frac{80 - 8.8}{83.2} = \frac{71.2}{83.2} = 0.856$$

b) Microconstituents at T_3 Proeutectic (primary) β of 92 wt% Cu, and **eutectic** of 28.1 wt% Cu overall:

$$W_{\beta p} = \frac{C_0 - C_e}{C_\beta - C_e} = \frac{80 - 28.1}{92 - 28.1} = \frac{51.9}{63.9} = 0.812 \qquad W_E = \frac{C_\beta - C_0}{C_\beta - C_e} = \frac{12}{63.9} = 0.188$$

Phases inside the eutectic:

$$W_{\beta,E} = W_\beta(T_3) - W_{\beta p} = 0.856 - 0.812 = 0.044 \qquad W_{\alpha,E} = W_\alpha(T_3) = 0.144$$

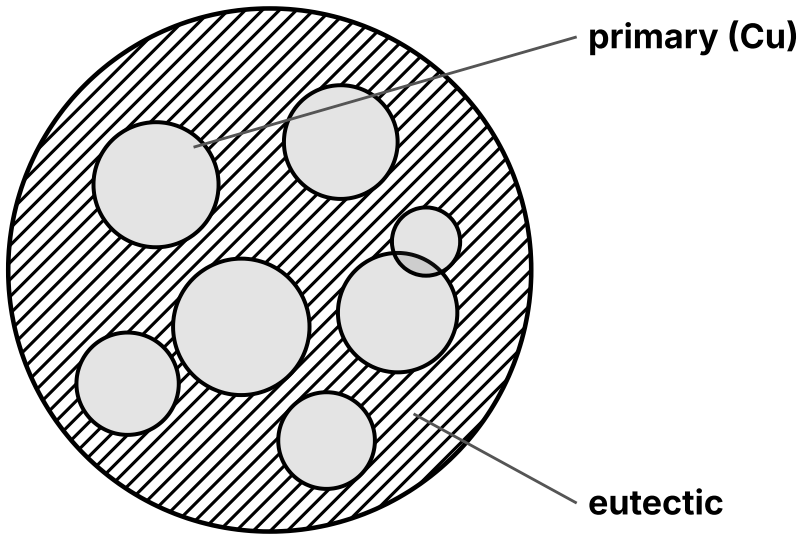


Figure 10. Ag-80 wt% Cu below 779 °C: primary (Cu) (81 %) in the eutectic (19 %).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- $W_{\alpha,E} + W_{\beta,E} = 0.144 + 0.044 = 0.188 = W_E$.
- Inside the eutectic α is $0.144/0.188 = 0.768$, the make-up of the eutectic alloy of Problem 5.
- Mass balance at T_3: $0.1442 \times 8.8 + 0.8558 \times 92 = 1.27 + 78.73 = 80.0$ wt% Cu.

Interpretation Between 900 °C and the eutectic temperature the amount of primary (Cu) grows from 63 % to 81 %, and the liquid becomes poorer in Cu, from 58 to 28.1 wt%. The final structure is mostly primary (Cu) crystals with about 19 % eutectic between them.

Check against the official solution

All values agree with the key. The worked page (Solution problems Phase diagrams I.pdf, pp. 11-12) adds a fourth temperature, $T_4 = 500$ °C, that is not in the statement of the sheet, and writes $\chi_{Cu,E} = 0.86 - 0.81 = 0.04$ (exactly $0.856 - 0.812 = 0.044$).

Problem 7 | Problems T4_PhaseDiagrams I.pdf #7 | Ag-Cu · hypoeutectic

The Hypoeutectic Ag-10 wt% Cu Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 2

7. Using the Cu-Ag phase diagram, for an alloy with composition C_0 = 10% Cu, for the temperatures $T_1 = 900$ °C, $T_2 = (780 + \Delta T)$ °C, $T_3 = (780 - \Delta T)$ °C, determine: a. Present phases, chemical composition of phases and mass fraction. b. Microconstituents, chemical composition and mass fraction at T_3.

Official Solution: a) T_1: Liquid (10%Cu); T_2: C_L = 28.1%Cu; C_α = 8.8%Cu; $X_\alpha = 0.94$; $X_L = 0.06$; T_3: C_α = 8.8%Cu; C_β = 92%Cu; $X_\alpha = 0.98$; $X_\beta = 0.02$. b) eutectic microconstituent ($X_E = 0.06$) and proeutectic α ($X_{\alpha p} = 0.94$)

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium cooling**, Ag-Cu diagram of the problem sheet (p. 2): eutectic 28.1 wt% Cu at 779.1 °C ≈ 780 °C; α = (Ag) of 8.8 and β = (Cu) of 92 wt% Cu.
2. **Reading of the figure at 900 °C:** the Ag-side liquidus falls from 961.93 °C (0 % Cu) to 779.1 °C (28.1 % Cu), and at 10 wt% Cu it is just below 900 °C, so at 900 °C the alloy is still liquid (as the official solution states).
3. $8.8 < C_0 = 10 < 28.1$: a **hypoeutectic** alloy, very close to the α limit; the primary phase is α [Session 7 Slide 30].

Data

Temperature	Region	Phase compositions (wt% Cu)
$T_1 = 900$ °C	L	$C_L = 10$
$T_2 = 780 + \Delta T$	α + L	$C_\alpha = 8.8$, $C_L = 28.1$
$T_3 = 780 - \Delta T$	α + β	$C_\alpha = 8.8$, $C_\beta = 92$

PHASE 2 · FRAMES & STRATEGY

Frame: the Ag-Cu diagram (wt% Cu, °C), one tie line per temperature; no change of basis. Lever rule for the phases [Session 7 Slide 16] and for the microconstituents (short tie line from α to the eutectic liquid) [Session 7 Slide 32].

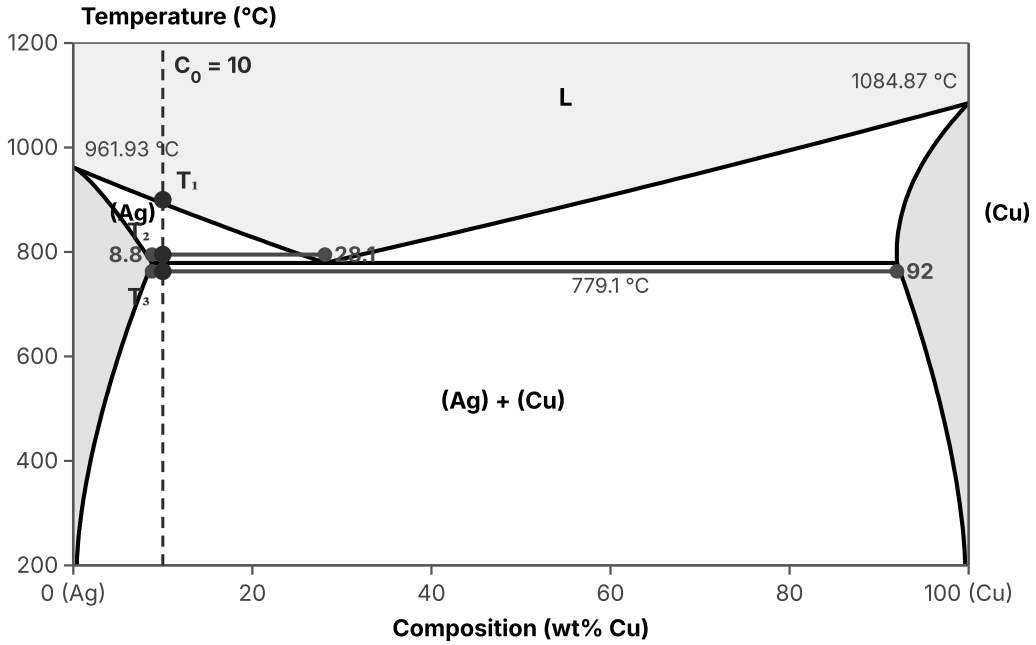


Figure 11. Ag-10 wt% Cu at T_1 = 900 °C (liquid, just above the liquidus, a reading of the figure), T_2 = 780 + ΔT and T_3 = 780 − ΔT. Problem sheet diagram.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases $T_1 = 900$ °C. Liquid, $C_L = 10$ wt% Cu, $W_L = 1$.

$T_2 = 780 + \Delta T$ (tie line length $28.1 - 8.8 = 19.3$):

$$W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{28.1 - 10}{28.1 - 8.8} = \frac{18.1}{19.3} = 0.938 \qquad W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{10 - 8.8}{19.3} = \frac{1.2}{19.3} = 0.062$$

$T_3 = 780 - \Delta T$ (length $92 - 8.8 = 83.2$):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{92 - 10}{92 - 8.8} = \frac{82}{83.2} = 0.986 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{10 - 8.8}{83.2} = \frac{1.2}{83.2} = 0.014$$

b) Microconstituents at T_3 Proeutectic (primary) α of 8.8 wt% Cu and **eutectic** of 28.1 wt% Cu overall:

$$W_{\alpha p} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{18.1}{19.3} = 0.938 \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{1.2}{19.3} = 0.062$$

Phases inside the eutectic:

$$W_{\alpha,E} = W_\alpha(T_3) - W_{\alpha p} = 0.986 - 0.938 = 0.048 \qquad W_{\beta,E} = W_\beta(T_3) = 0.014$$

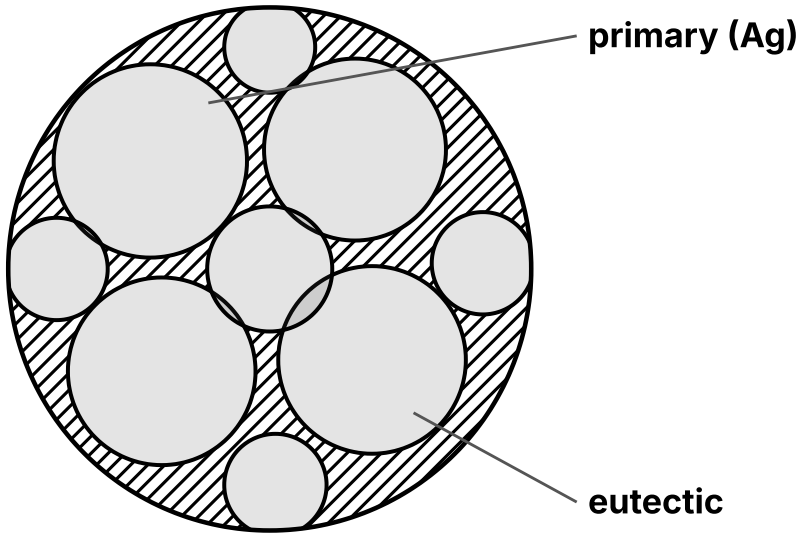


Figure 12. Ag-10 wt% Cu below 779 °C: primary (Ag) (94 %) with a little eutectic (6 %) between the grains.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- $W_{\alpha,E} + W_{\beta,E} = 0.048 + 0.014 = 0.062 = W_E$; inside the eutectic α is $0.048/0.062 = 0.77$, as in Problem 5.
- Mass balance at T_3: $0.9856 \times 8.8 + 0.0144 \times 92 = 8.67 + 1.33 = 10.0$ wt% Cu.

Interpretation The alloy is only 1.2 wt% Cu beyond the solubility limit of α, so almost all of it is primary (Ag) and just 6 % is eutectic, found as thin films between the α grains. An alloy below 8.8 wt% Cu would form no eutectic at all.

Check against the official solution

At T_1, T_2 and in part (b) the results agree with the key.

DISCREPANCY WITH THE OFFICIAL SOLUTION

At T_3 the key gives $X_\alpha = 0.98$ and $X_\beta = 0.02$. With the data of the problem, $W_\alpha = 82/83.2 = 0.9856$ and $W_\beta = 1.2/83.2 = 0.0144$, which round to 0.99 and 0.01. The computed values are kept; the key's 0.98 and 0.02 are not reproduced by any rounding of the data.

`Problem 8` `Problems T4_PhaseDiagrams I.pdf #8` `Diagram construction`

Drawing a Eutectic Diagram and the 40 % B Alloy

`Original statement · Problems T4_PhaseDiagrams I.pdf, p. 3`

8. Two metals (A and B) have partial solubility among each other. Metal A, with a melting point at 700 °C, can dissolve 5% of B at 0 °C, and its maximum solubility occurs at 400 °C, being 25% B. On the other hand, B metal melts at 600 °C, can dissolve 10% of A at 0 °C and its maximum solubility occurs at 400 °C, being 20% A. At 400 °C there is an eutectic reaction when the B content is 65%. a. Draw and identify the parts of the phase diagram in equilibrium (suppose straight lines). b. For an alloy 40% B, determine the phases composition and their relative proportions at 450 °C, 400 °C + ΔT and 400 °C – ΔT. c. For an alloy 40% B, draw its microstructure indicating the microconstituents and the mass fraction of each one. Additionally, indicate the mass fraction of each of the phases present in such microconstituents.

Official Solution: b) 450 °C: C_L **= 55%B;** C_α **= 20%B;** $X_\alpha = 0.43$**;** $X_L = 0.57$**; 400 °C + ΔT:** C_L **= 65%B;** C_α **= 25%B;** $X_\alpha = 0.62$**;** $X_L = 0.38$**; 400 °C - ΔT:** C_β **= 80%B;** C_α **= 25%B;** $X_\alpha = 0.73$ **;** $X_\beta = 0.27$**. c) Eutectic microconstituent and proeutectic α;** $X_{\alpha p} = 0.62$**;** $X_E = 0.38$**; Phases in the eutectic microconstituent:** $X(\alpha_E) = 0.11$**;** $X(\beta_E) = 0.27$**.**

`PHASE 1 · HYPOTHESES & PARAMETERS`

Hypotheses

1. **Binary eutectic system** with partial solid solubility: α (rich in A) and β (rich in B) [Session 7 Slide 26].
2. **Straight lines** between the given points, as the statement asks.
3. Compositions in % B (mass); "20 % A" in β means 80 % B, and "10 % A" means 90 % B.

Points of the diagram

Point	Composition (% B)	Temperature (°C)	Meaning
Melting point of A	0	700	top of α solidus and of the A-side liquidus
Melting point of B	100	600	top of β solidus and of the B-side liquidus
Eutectic point E	65	400	lowest point of the liquid
Max. solubility of B in α	25	400	end of the eutectic line, α side
Max. solubility of A in β	80	400	end of the eutectic line, β side
α solvus at 0 °C	5	0	

Point	Composition (% B)	Temperature (°C)	Meaning
β solvus at 0 °C	90	0	

PHASE 2 · FRAMES & STRATEGY

Frame: composition (% B) on the horizontal axis and temperature (°C) on the vertical one. Each boundary is a straight segment, so a composition on it at a temperature T comes from linear interpolation between its two end points; the tie lines are horizontal. No change of basis is needed.

Lines of the diagram (part a):

- liquidus: $(0, 700) \to (65, 400)$ and $(100, 600) \to (65, 400)$;
- solidus: $(0, 700) \to (25, 400)$ for α and $(100, 600) \to (80, 400)$ for β;
- eutectic line: $(25, 400) \to (80, 400)$;
- solvus: $(25, 400) \to (5, 0)$ for α and $(80, 400) \to (90, 0)$ for β.

Regions: L above the liquidus; α + L and β + L between liquidus and solidus; α and β next to the axes, bounded by solidus and solvus; α + β below the eutectic line, between the solvus lines.

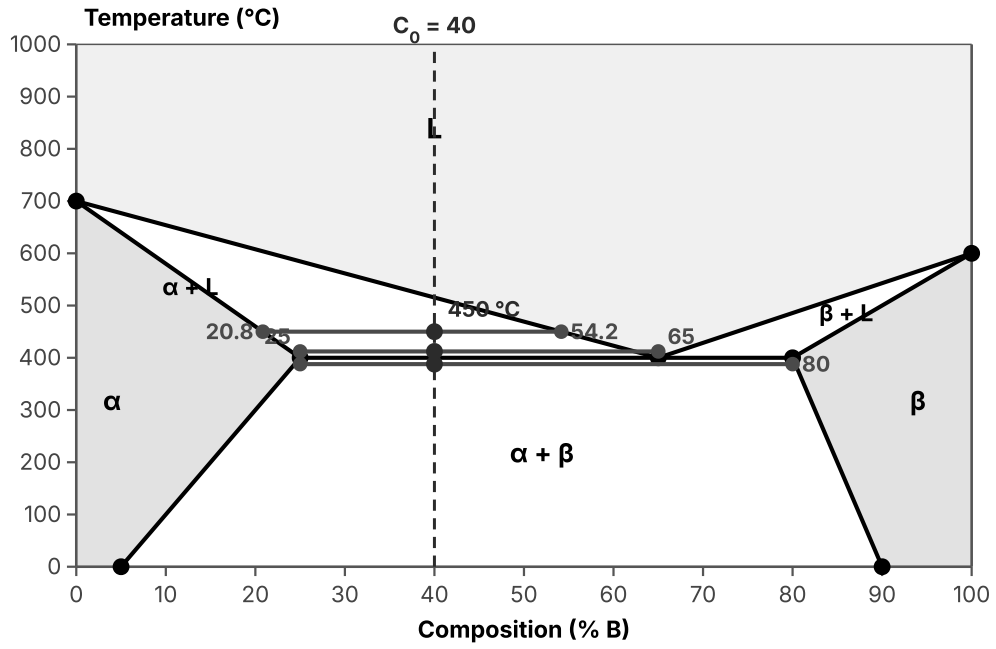


Figure 13. A-B diagram with straight lines: melting points 700 °C (A) and 600 °C (B), eutectic at 400 °C and 65 % B, maximum solubilities 25 % B in α and 20 % A in β (80 % B) at 400 °C, and 5 % B in α and 10 % A in β (90 % B) at 0 °C. Dashed: the 40 % B alloy, with its tie lines at 450 °C and just above and below 400 °C.

PHASE 3 · STEP-BY-STEP DERIVATION

b) Phases of the 40 % B alloy 450 °C. The alloy is between the A-side liquidus and the α solidus: **α + L.** On a straight segment from $(0, 700)$ to $(C_1, 400)$ the composition at T is $C = C_1 \dfrac{700 - T}{700 - 400}$:

$$C_L = 65 \times \frac{700 - 450}{300} = 65 \times \frac{250}{300} = 54.17\ \%\text{B} \qquad C_\alpha = 25 \times \frac{250}{300} = 20.83\ \%\text{B}$$

$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha} = \frac{40 - 20.83}{54.17 - 20.83} = \frac{19.17}{33.33} = 0.575 \qquad W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha} = \frac{54.17 - 40}{33.33} = \frac{14.17}{33.33} = 0.425$$

400 °C + ΔT. Still α + L, with the ends of the tie line at the eutectic temperature: $C_\alpha = 25$, $C_L = 65$ % B (length 40):

$$W_L = \frac{40 - 25}{65 - 25} = \frac{15}{40} = 0.375 \qquad W_\alpha = \frac{65 - 40}{40} = \frac{25}{40} = 0.625$$

400 °C – ΔT. α + β; $C_\alpha = 25$, $C_\beta = 80$ % B (length 55):

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{80 - 40}{80 - 25} = \frac{40}{55} = 0.727 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{40 - 25}{55} = \frac{15}{55} = 0.273$$

c) Microconstituents at 400 °C – ΔT $25 < 40 < 65$: hypoeutectic, so the microconstituents are **proeutectic α** (25 % B) and **eutectic** (65 % B overall). The α present just above 400 °C is the proeutectic α, and the liquid becomes eutectic [Session 7 Slide 32]:

$$W_{\alpha p} = \frac{C_e - C_0}{C_e - C_\alpha} = \frac{65 - 40}{65 - 25} = 0.625 \qquad W_E = \frac{C_0 - C_\alpha}{C_e - C_\alpha} = \frac{40 - 25}{40} = 0.375$$

Phases inside the eutectic:

$$W_{\alpha,E} = W_\alpha - W_{\alpha p} = 0.7273 - 0.6250 = 0.1023 \qquad W_{\beta,E} = W_\beta = 0.273$$

The same numbers from the make-up of the eutectic, which is $(80-65)/(80-25) = 15/55 = 0.273$ α and $40/55 = 0.727$ β: $W_{\alpha,E} = 0.375 \times 0.273 = 0.102$ and $W_{\beta,E} = 0.375 \times 0.727 = 0.273$.

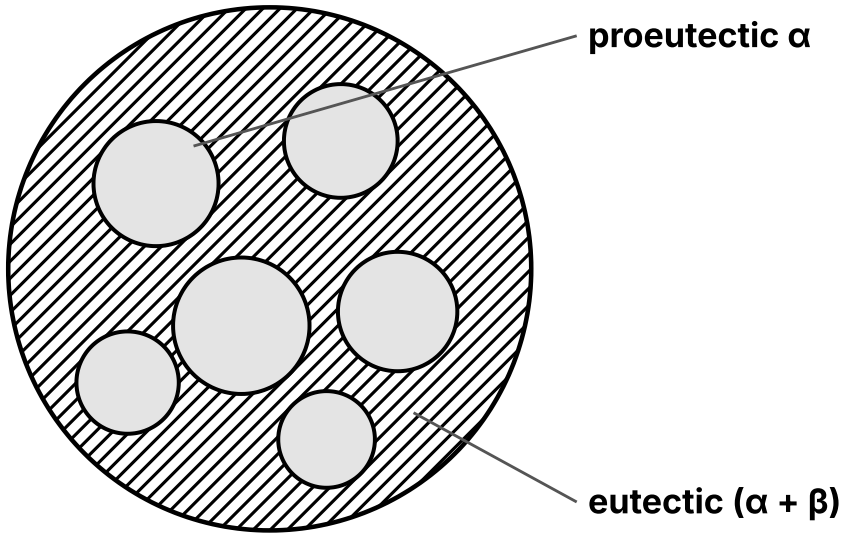


Figure 14. 40 % B alloy at 400 − ΔT: proeutectic α (62.5 %) in the eutectic α + β (37.5 %).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- Mass balance at 450 °C: $0.425 \times 20.83 + 0.575 \times 54.17 = 8.85 + 31.15 = 40.0$ % B.
- Mass balance at 400 °C − ΔT: $0.727 \times 25 + 0.273 \times 80 = 18.18 + 21.82 = 40.0$ % B.
- $W_{\alpha,E} + W_{\beta,E} = 0.102 + 0.273 = 0.375 = W_E$, and $W_{\alpha p} + W_{\alpha,E} = 0.625 + 0.102 = 0.727 = W_\alpha$.

Interpretation On cooling from the liquid, α starts to form at the liquidus ($700 - 300 \times 40/65 = 515$ °C for this alloy), grows to 62.5 % at the eutectic temperature, and the remaining 37.5 % of liquid, of 65 % B, becomes eutectic at 400 °C. Most of the eutectic is β (0.273 of the alloy), which is all the β of the alloy.

Check against the official solution

At 400 °C ± ΔT the results agree with the key (0.62/0.38 are 0.625/0.375; 0.73/0.27). The key writes "C_α = 25%Cu" at 400 °C + ΔT, a typo for 25 % B.

DISCREPANCY WITH THE OFFICIAL SOLUTION

450 °C. The key reads $C_L = 55$ and $C_\alpha = 20$ % B from its drawing and gives $X_L = 0.57$, $X_\alpha = 0.43$. With the straight lines that the statement requires, linear interpolation gives $C_L = 54.17$ and $C_\alpha = 20.83$ % B, hence $W_L = 0.575$ and $W_\alpha = 0.425$. The difference comes from reading the drawing; the interpolated values are kept.

α in the eutectic. The key gives $X(\alpha_E) = 0.11$, which is $0.73 - 0.62$ with rounded values. Exactly, $0.7273 - 0.6250 = 0.1023$, confirmed by $0.375 \times 15/55 = 0.102$; the key's 0.11 is a rounding effect.

Problem 9 | Problems T4_PhaseDiagrams I.pdf #9 | Al-Ni · intermetallics

Invariant Reactions and an Al-20 wt% Ni Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 3

9. Given the Al-Ni phase diagram, answer the questions below. a. What is the maximum solubility of aluminum in solid state nickel? What about nickel in aluminum? b. Write down the invariant reactions, indicating the temperature, the reaction in equation form, the composition of each phase and the name of the reaction. c. Given an Al-20% Ni alloy, indicate at 400 °C: composition and mass fraction of the phases; composition and mass fraction of the microconstituents; draw the microstructure that can be expected when cooling in equilibrium conditions. Point out each microconstituent with arrows.

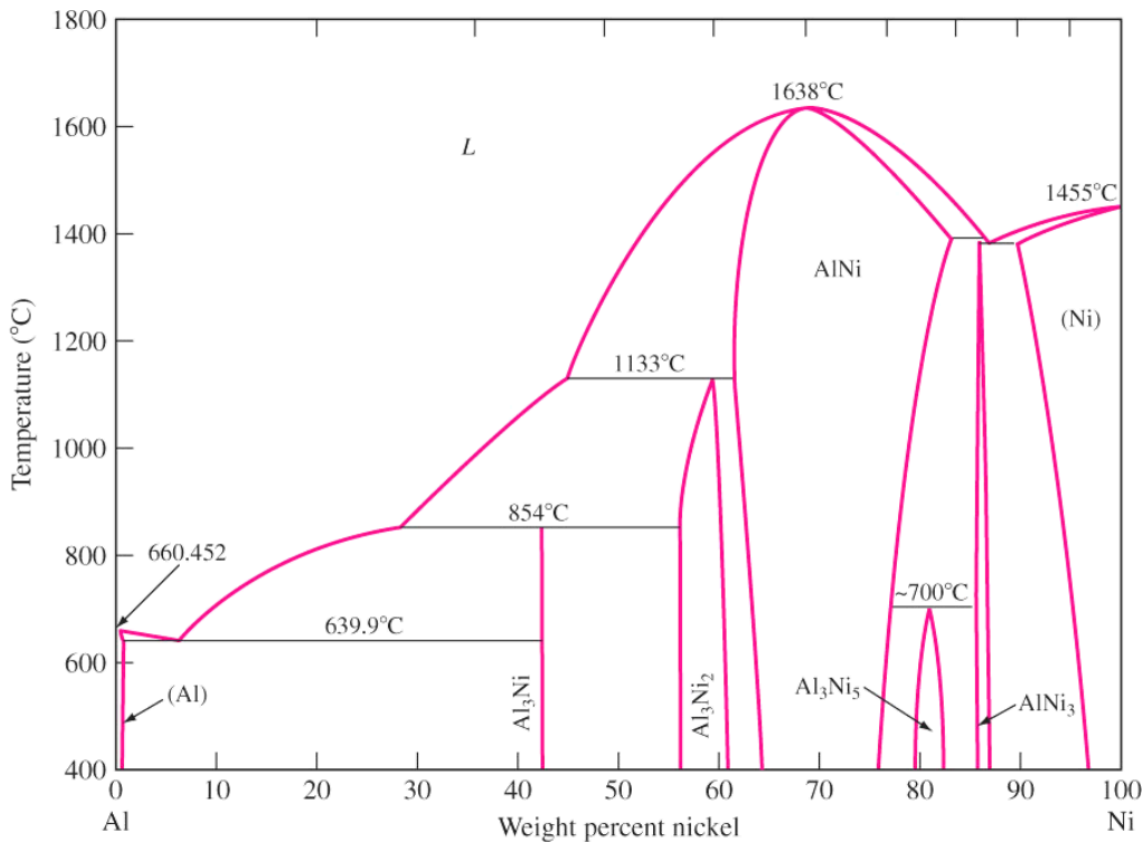


Figure 16. Al-Ni phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 3).

Official Solution: a) 10%Al; 1%Ni. b) eutectic reaction (639 °C; 1380 °C); peritectoid reaction (700 °C); peritectic reaction (854 °C; 1133 °C, 1400 °C). c) C_{Al_3Ni} = 43%Cu; C_α = 1%Cu; X_α = 0.55; X_{Al_3Ni} = 0.45; Microconstituents: eutectic (X = 0.64) and proeutectic Al_3Ni (X = 0.36)

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium** and the Al-Ni diagram of the sheet (weight percent nickel). Every composition and temperature below is a **reading of the figure**, to about ±1 wt% and ±10 °C, except the temperatures printed on it (639.9, 854, 1133, ~700, 1638 °C).
2. **Invariant reactions** are the horizontal lines where three phases meet ($F = 2 - 3 + 1 = 0$) [Session 7 Slide 13; Session 8 Slide 6]; their type follows from the shape around the line (V above: eutectic; inverted V below: peritectic or peritectoid) [Session 8 Slide 6].
3. **Al3Ni** is drawn as a vertical line (stoichiometric compound) at about 42 wt% Ni [Session 8 Slide 15].
4. The (Al) phase is the very thin field along the left axis; its solubility limit is about 1 wt% Ni at most.

Data read on the diagram

Point	Reading
Eutectic at 639.9 °C	L ≈ 6 wt% Ni, (Al) ≈ 1 wt% Ni, Al_3Ni ≈ 42 wt% Ni
(Al) at 400 °C	≈ 1 wt% Ni
(Ni) at its widest	≈ 90 wt% Ni, i.e. ≈ 10 wt% Al

PHASE 2 · FRAMES & STRATEGY

Frame: the Al-Ni diagram (wt% Ni, °C). Each question is answered on horizontal lines (tie lines and invariant lines), so the work is one-dimensional and no change of basis is needed.

Strategy. (a) Maximum solubility = the widest point of each terminal solid solution. (b) Scan the diagram from the bottom up for horizontal lines and name each reaction by its shape. (c) Lever rule on the (Al)-Al3Ni tie line for the phases, and on the short tie line from the eutectic liquid to Al3Ni (just above 639.9 °C) for the microconstituents [Session 7 Slide 32].

PHASE 3 · STEP-BY-STEP DERIVATION

a) Maximum solubilities

- **Al in solid Ni:** the (Ni) field is widest at its horizontal line near 1385 °C, at about 90 wt% Ni: about **10 wt% Al**.
- **Ni in solid Al:** the (Al) field is widest at the eutectic temperature, 639.9 °C, at about **1 wt% Ni** (the field is too thin to read more precisely).

b) Invariant reactions (on cooling)

T (°C)	Reaction	Name
639.9	$L\,(6) \rightarrow (\mathrm{Al})\,(1) + \mathrm{Al_3Ni}\,(42)$	eutectic
~700	$\mathrm{AlNi}\,(76) + \mathrm{AlNi_3}\,(86) \rightarrow \mathrm{Al_3Ni_5}\,(81)$	peritectoid
854	$L\,(28) + \mathrm{Al_3Ni_2}\,(56) \rightarrow \mathrm{Al_3Ni}\,(42)$	peritectic
1133	$L\,(44) + \mathrm{AlNi}\,(61) \rightarrow \mathrm{Al_3Ni_2}\,(59)$	peritectic
≈1395	$L\,(86.5) + \mathrm{AlNi}\,(83) \rightarrow \mathrm{AlNi_3}\,(86)$	peritectic
≈1385	$L\,(86.5) \rightarrow \mathrm{AlNi_3}\,(86) + (\mathrm{Ni})\,(90)$	eutectic

(compositions in wt% Ni, readings of the figure). The two reactions near 1390 °C are only a few degrees and a few percent apart; the key gives them as 1400 °C (peritectic) and 1380 °C (eutectic).

c) Al-20 wt% Ni at 400 °C Phases. (Al) of about 1 wt% Ni and Al3Ni of 42 wt% Ni (tie line length $42 - 1 = 41$):

$$W_{(\mathrm{Al})} = \frac{C_{\mathrm{Al_3Ni}} - C_0}{C_{\mathrm{Al_3Ni}} - C_{(\mathrm{Al})}} = \frac{42 - 20}{42 - 1} = \frac{22}{41} = 0.537 \qquad W_{\mathrm{Al_3Ni}} = \frac{20 - 1}{41} = \frac{19}{41} = 0.463$$

Microconstituents. $6 < 20 < 42$: the alloy is **hypereutectic** with respect to the 639.9 °C eutectic, so Al3Ni forms first (primary, proeutectic) and the remaining liquid of 6 wt% Ni becomes eutectic. Just above 639.9 °C (tie line from 6 to 42, length 36):

$$W_{\mathrm{Al_3Ni},p} = \frac{C_0 - C_e}{C_{\mathrm{Al_3Ni}} - C_e} = \frac{20 - 6}{42 - 6} = \frac{14}{36} = 0.389 \qquad W_E = \frac{42 - 20}{36} = \frac{22}{36} = 0.611$$

The microconstituents keep these amounts on cooling to 400 °C (Al3Ni has a fixed composition and the (Al) solvus barely moves).

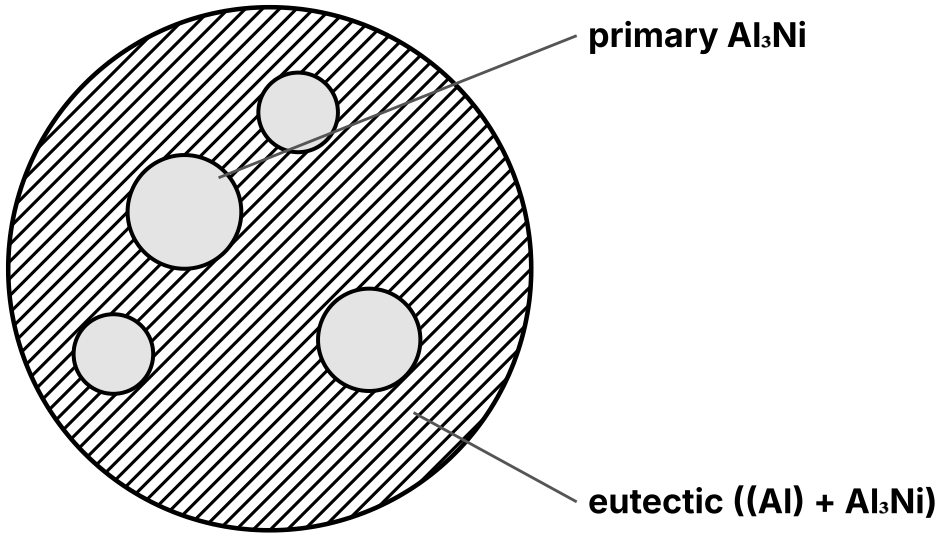


Figure 15. Al-20 wt% Ni at 400 °C: primary Al_3Ni crystals (about 39 %) in the eutectic (Al) + Al_3Ni (about 61 %).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- Mass balance at 400 °C: $0.537 \times 1 + 0.463 \times 42 = 0.54 + 19.45 = 20.0$ wt% Ni.
- All the (Al) is inside the eutectic: $W_{(\mathrm{Al})} = 0.537 < W_E = 0.611$, and the rest of the eutectic, $0.611 - 0.537 = 0.074$, is eutectic Al_3Ni; total $Al_3Ni = 0.389 + 0.074 = 0.463$.

Interpretation The alloy is mostly eutectic of soft (Al) with about 39 % of hard Al_3Ni crystals that grew from the liquid between about 800 °C (liquidus at 20 wt% Ni, reading) and 639.9 °C. The Ni-rich side of the diagram is crowded with intermetallic compounds formed by peritectic and peritectoid reactions.

Check against the official solution

Parts (a) and (b) agree with the key (10 % Al, 1 % Ni; eutectics at 639.9 and ~1380 °C, peritectoid at ~700 °C, peritectics at 854, 1133 and ~1400 °C). The key writes the compositions of part (c) as "%Cu"; they are wt% Ni.

DISCREPANCY WITH THE OFFICIAL SOLUTION

For part (c) the key uses $C_{\mathrm{Al_3Ni}} = 43$ % and a eutectic liquid of about 7 % Ni, and gives $X_\alpha = 0.55$, $X_{\mathrm{Al_3Ni}} = 0.45$, eutectic 0.64 and primary Al_3Ni 0.36. The figure places the Al_3Ni line at about 42 % and the eutectic point at about 6 % Ni, which give 0.537/0.463 and 0.611/0.389. With the key's readings the same formulas give $(43 - 20)/(43 - 1) = 23/42 = 0.548$ for (Al) and $(43 - 20)/(43 - 7) = 23/36 = 0.64$ for the eutectic: the difference comes only from the readings, which are within the precision of the drawing. The values read here are kept.

Problem 10 | Problems T4_PhaseDiagrams I.pdf #10 | Cu-Sn · eutectoids

Reactions and Eutectoids in the Cu-Sn System

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 4

10. Considering the Cu-Sn phase diagram: a. Indicate the type of transformations, and write the reactions that take place at T_1 = 798 °C; T_2 = 755 °C; T_3 = 588 °C; T_4 = 415 °C; T_5 = 1084.87 °C. b. For a Cu-20%Sn alloy, indicate the phases and microconstituents, their composition and amount at T_1: 521 °C and T_2: 519 °C. c. Determine the percentage of Sn in an alloy having 54% liquid phase at 900 °C. Is this alloy in solid state at 700 °C?

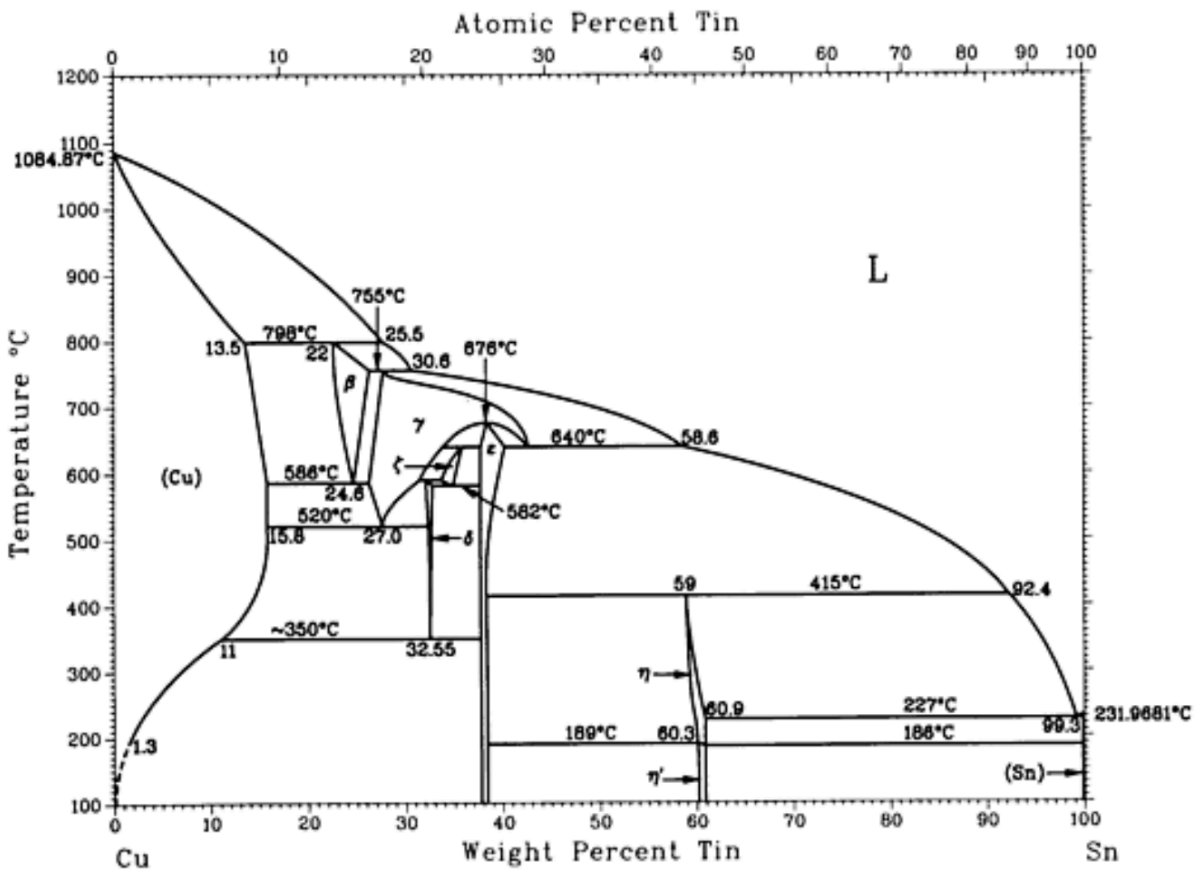


Figure 17. Cu-Sn phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 4).

Official Solution: a) peritectic; peritectic, eutectoid; peritectic and solidification of Cu. b) 521 °C: $C_{(Cu)}$ = 15.8%Sn; C_γ = 27%Sn; X_γ = 0.37; $X_{(Cu)}$ = 0.63; X(eutectoid): 0.48; X(proeutectoid Cu): 0.52. 519 °C: $C_{(Cu)}$ = 15.8%Sn; C_δ = 32%Sn; X_δ = 0.26; $X_{(Cu)}$ = 0.74; X(eutectoid): 0.37 and X(proeutectoid Cu): 0.63. c) 14.5%Sn

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**, Cu-Sn diagram of the sheet (wt% Sn). Printed values: 798 °C with 13.5, 22 and 25.5 wt% Sn; 755 °C with 30.6; 586 °C with 24.6; 520 °C with 15.8 and 27.0; 415 °C with 59 and 92.4. The rest are **readings of the figure** (±1 wt%).
2. The statement's 588 °C is the 586 °C line of the diagram.
3. The type of each reaction follows from its shape [Session 8 Slide 6].

Data

Line	Phases and compositions (wt% Sn)
798 °C	(Cu) 13.5, β 22, L 25.5
755 °C	β ≈ 24, γ ≈ 25.5, L 30.6
586 °C	(Cu) 15.8, β 24.6, γ ≈ 25.5
520 °C	(Cu) 15.8, γ 27.0, δ ≈ 32
415 °C	ε ≈ 38, η 59, L 92.4
900 °C	(Cu) ≈ 9, L ≈ 19 (on the solidus and liquidus)

PHASE 2 · FRAMES & STRATEGY

Frame: the Cu-Sn diagram (wt% Sn, °C); all the work is on horizontal lines, so no change of basis is needed.

Strategy. (a) Name each line by its shape. (b) Lever rule on the tie lines just above and just below 520 °C; the microconstituents come from the history of the alloy (it crosses 798, 586 and 520 °C). (c) Write the lever rule for the liquid at 900 °C and solve it for C_0.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Transformations

T	Reaction on cooling	Type
T_1 = 798 °C	$(\mathrm{Cu})\,(13.5) + L\,(25.5) \rightarrow \beta\,(22)$	peritectic
T_2 = 755 °C	$\beta\,(\approx 24) + L\,(30.6) \rightarrow \gamma\,(\approx 25.5)$	peritectic
T_3 = 586 °C	$\beta\,(24.6) \rightarrow (\mathrm{Cu})\,(15.8) + \gamma\,(\approx 25.5)$	eutectoid
T_4 = 415 °C	$\varepsilon\,(\approx 38) + L\,(92.4) \rightarrow \eta\,(59)$	peritectic
T_5 = 1084.87 °C	$L \rightarrow (\mathrm{Cu})$ for pure Cu	solidification of a pure metal

b) Cu-20 wt% Sn History. $13.5 < 20 < 22$: at 798 °C the peritectic consumes all the liquid and leaves (Cu) + β. On cooling, β is transformed at 586 °C by the eutectoid into (Cu) + γ, and γ at 520 °C by a second eutectoid into (Cu) + δ.

Just above 586 °C ((Cu) 15.8, β 24.6; length 8.8):

$$W_\beta = \frac{20 - 15.8}{24.6 - 15.8} = \frac{4.2}{8.8} = 0.477 \qquad W_{(\mathrm{Cu})} = \frac{24.6 - 20}{8.8} = \frac{4.6}{8.8} = 0.523$$

This (Cu) is **proeutectoid**, and this β becomes the **eutectoid** microconstituent at 586 °C.

At 521 °C (phases), (Cu) 15.8 and γ 27.0 (length 11.2):

$$W_\gamma = \frac{20 - 15.8}{27.0 - 15.8} = \frac{4.2}{11.2} = 0.375 \qquad W_{(\mathrm{Cu})} = \frac{27.0 - 20}{11.2} = \frac{7.0}{11.2} = 0.625$$

Microconstituents at 521 °C: proeutectoid (Cu) 0.523 and eutectoid [(Cu) + γ] 0.477.

At 519 °C (phases), below the 520 °C eutectoid $\gamma\,(27.0) \rightarrow (\mathrm{Cu})\,(15.8) + \delta\,(32)$; (Cu) 15.8 and δ ≈ 32 (length 16.2):

$$W_\delta = \frac{20 - 15.8}{32 - 15.8} = \frac{4.2}{16.2} = 0.259 \qquad W_{(\mathrm{Cu})} = \frac{32 - 20}{16.2} = \frac{12}{16.2} = 0.741$$

Microconstituents at 519 °C, counted with respect to the 520 °C reaction: the γ present at 521 °C (0.375) becomes eutectoid [(Cu) + δ], and all the (Cu) that existed before (0.625) counts as proeutectoid. That (Cu) is made of the proeutectoid (Cu) of 586 °C (0.523) plus the (Cu) of the first eutectoid ($0.625 - 0.523 = 0.102$).

c) Alloy with 54 % liquid at 900 °C At 900 °C the tie line goes from (Cu) at $C_S \approx 9$ to L at $C_L \approx 19$ wt% Sn (readings). The lever rule for the liquid:

$$W_L = \frac{C_0 - C_S}{C_L - C_S} \;\Rightarrow\; C_0 = C_S + W_L\,(C_L - C_S) = 9 + 0.54 \times (19 - 9) = 9 + 5.4 = 14.4 \approx 14.5\ \mathrm{wt\%\ Sn}$$

Solid at 700 °C? $13.5 < 14.5 < 22$, so at 798 °C the peritectic uses up all the liquid: below 798 °C the alloy is (Cu) + β, both solid. At 700 °C the (Cu) boundary is at about 14.5 wt% Sn (it goes from 13.5 at 798 °C to 15.8 at 586 °C), so the alloy is (Cu), practically single-phase. **Yes, it is solid at 700 °C.**

PHASE 4 · CHECKS & INTERPRETATION

Checks

- 519 °C: $0.741 \times 15.8 + 0.259 \times 32 = 11.71 + 8.29 = 20.0$ wt% Sn.
- 521 °C: $0.625 \times 15.8 + 0.375 \times 27.0 = 9.88 + 10.13 = 20.0$ wt% Sn.
- (c): with $C_0 = 14.4$, $W_L = (14.4 - 9)/(19 - 9) = 0.54$.

Interpretation The Cu-rich side of Cu-Sn (bronzes) shows how one alloy can cross several invariant reactions: a peritectic during solidification and two eutectoids in the solid. Each eutectoid turns a single solid phase into a fine two-phase mixture.

Check against the official solution

All values agree with the key: 0.37/0.63 and 0.48/0.52 at 521 °C; 0.26/0.74 and 0.37/0.63 at 519 °C (the key counts microconstituents with respect to the 520 °C reaction, as explained in part b); 14.5 % Sn. The key's part (c) does not answer the question about 700 °C.

> ERRATUM IN THE WORKED SOLUTION
>
> The annotated diagram of Solution problems Phase diagrams I.pdf, p. 35, labels the field above the liquidus at about 50 % Sn and 900 °C as "(Cu)+ε"; that field is the liquid L, as the sheet's own diagram shows.

Problem 11 | Problems T4_PhaseDiagrams I.pdf #11 | Diagram construction

Drawing a Diagram with a Stoichiometric Compound

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 4

11. Draw the hypothetic phase diagram of A and B from the following data. a) Melting temperatures: A = 850 °C; B = 350 °C. b) A and B form an stoichiometric compound AB2 that melts at 1000 °C; neither A nor B are soluble in the compound. c) There is an invariant eutectic point (70% at. of A, 700 °C). d) There is an invariant peritectic point (80% at. of B, 500 °C). e) The composition of the liquid at 500 °C is 95% at. of B. f) The solubility of B in A is 10% at. at 700 °C and 2% at. at 25 °C. g) The solubility of A in B is 25 % at. at 25 °C. Connect the points with straight lines. Indicate all the phase areas and points in the diagram, and write the reactions of the three phases.

Official Solution: not given in the sheet

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. Compositions in **atomic percent of B** (at.% B) on the horizontal axis; straight lines between the given points, as the statement asks.
2. **AB2** contains two B atoms per A atom: $x_B = 2/3 = 66.7$ at.% B. It is stoichiometric and dissolves neither A nor B, so it is a **vertical line** [Session 8 Slide 15]; it melts at 1000 °C as a single phase, so it melts **congruently** (a maximum of the liquidus) [Session 8 Slide 17].
3. The terminal solid solutions are α (B dissolved in A) and β (A dissolved in B).
4. "80 % at. of B" for the peritectic point is the composition of the solid β formed there; the liquid at 500 °C has 95 at.% B (point e).

Points

Point	at.% B	T (°C)
Melting point of A	0	850
Melting point of B	100	350
AB_2 (congruent melting)	66.7	1000
Eutectic E (liquid)	30	700
α at E (max. solubility of B in A)	10	700
α at 25 °C	2	25
Peritectic P (β formed)	80	500
Liquid at 500 °C	95	500

Point	at.% B	T (°C)
β at 25 °C (25 at.% A)	75	25

PHASE 2 · FRAMES & STRATEGY

Frame: composition (at.% B) against temperature (°C). The diagram is built from straight segments between the points; no change of basis is involved.

How the points connect.

- Left part (A to AB_2): a **eutectic** system between α and AB_2 [Session 7 Slide 26]: liquidus from A (0, 850) to E (30, 700) and from E up to AB_2 (66.7, 1000); α solidus from (0, 850) to (10, 700); eutectic line from α (10) to AB_2 (66.7) at 700 °C; α solvus from (10, 700) to (2, 25).
- Right part (AB_2 to B): a **peritectic** system [Session 8 Slide 8]: liquidus from AB_2 (66.7, 1000) down to (95, 500) and then to B (100, 350); peritectic line from AB_2 (66.7) to L (95) at 500 °C, with β (80) on it; β solidus from (80, 500) to B (100, 350); β solvus from (80, 500) to (75, 25).

PHASE 3 · STEP-BY-STEP DERIVATION

The diagram

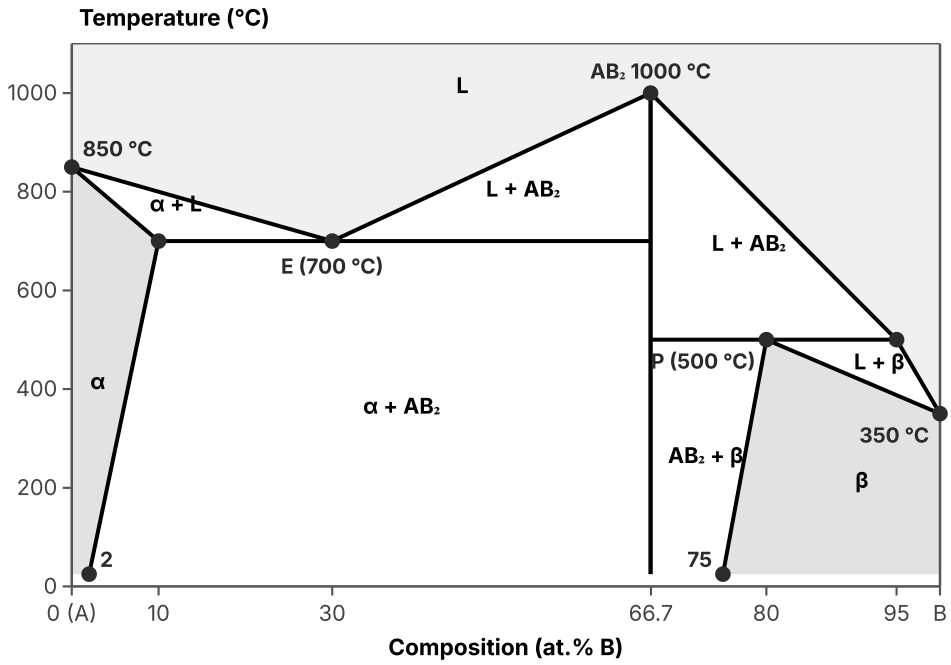


Figure 18. A-B with straight lines: A melts at 850 °C, B at 350 °C, the stoichiometric compound AB_2 (66.7 at.% B) melts congruently at 1000 °C; eutectic E at 700 °C and 30 at.% B; peritectic P at 500 °C (β of 80, liquid of 95 at.% B); solubility of B in α 10 at.% at 700 °C and 2 at.% at 25 °C; solubility of A in β 25 at.% at 25 °C.

Phase areas

Region	Bounded by
L	above the liquidus
α + L	0-30 at.% B, between 850 and 700 °C
α	left of the α solidus and solvus
α + AB_2	10-66.7 at.% B, below 700 °C
L + AB_2 (left)	30-66.7 at.% B, between 700 °C and the liquidus
L + AB_2 (right)	66.7-95 at.% B, between 500 °C and the liquidus
AB_2 + β	66.7 at.% B to the β solvus, below 500 °C
L + β	between the β solidus (80 → 100) and the liquidus (95 → 100), from 500 to 350 °C
β	right of the β solidus and solvus

Reactions of three phases (on cooling)

- **Eutectic, 700 °C:** $L\,(30\ \text{at.\%\ B}) \rightarrow \alpha\,(10\ \text{at.\%\ B}) + \text{AB}_2\,(66.7\ \text{at.\%\ B})$
- **Peritectic, 500 °C:** $\text{AB}_2\,(66.7\ \text{at.\%\ B}) + L\,(95\ \text{at.\%\ B}) \rightarrow \beta\,(80\ \text{at.\%\ B})$

AB2 melting at 1000 °C involves only two phases (AB2 and a liquid of the same composition), so it is not a three-phase reaction.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- Each three-phase line has the middle phase in the right place: at 700 °C the liquid (30) lies **between** the two solids (10 and 66.7), the shape of a eutectic; at 500 °C the new solid β (80) lies between AB2 (66.7) and L (95), with the liquid at one end, the shape of a peritectic [Session 8 Slide 6].
- Phase rule: on each horizontal line three phases coexist, $F = 2 - 3 + 1 = 0$; in each two-phase area $F = 1$ [Session 7 Slide 13].

Interpretation The compound AB2 splits the diagram into two simpler diagrams, A-AB2 (eutectic) and AB2-B (peritectic). Since B melts at only 350 °C, the liquid persists to low temperatures on the B side, and β forms from AB2 and the B-rich liquid at 500 °C.

Check against the official solution

The sheet gives no solution for this problem and the official solutions PDF has no page for it, so there is nothing to compare with.

Problem 12 | Problems T4_PhaseDiagrams I.pdf #12 | Ni-Mo · hypereutectic

Invariant Reactions and a Ni-40 wt% Mo Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 5

12. According to the Ni-Mo phase, indicate: a. The equations that represent the invariant reactions, specifying the composition of the phases, temperature and type of reaction. b. The maximum solubility of Mo in Ni and at what temperature it is produced. c. The maximum solubility of Ni in Mo and at what temperature it is produced. d. Melting temperature of Ni and Mo. e. Present phases, their composition and amount for a Ni-40%Mo alloy at 1309 °C. f. Microconstituents and their amount for a Ni-40%Mo alloy at 1309 °C. Sketch the microstructure.

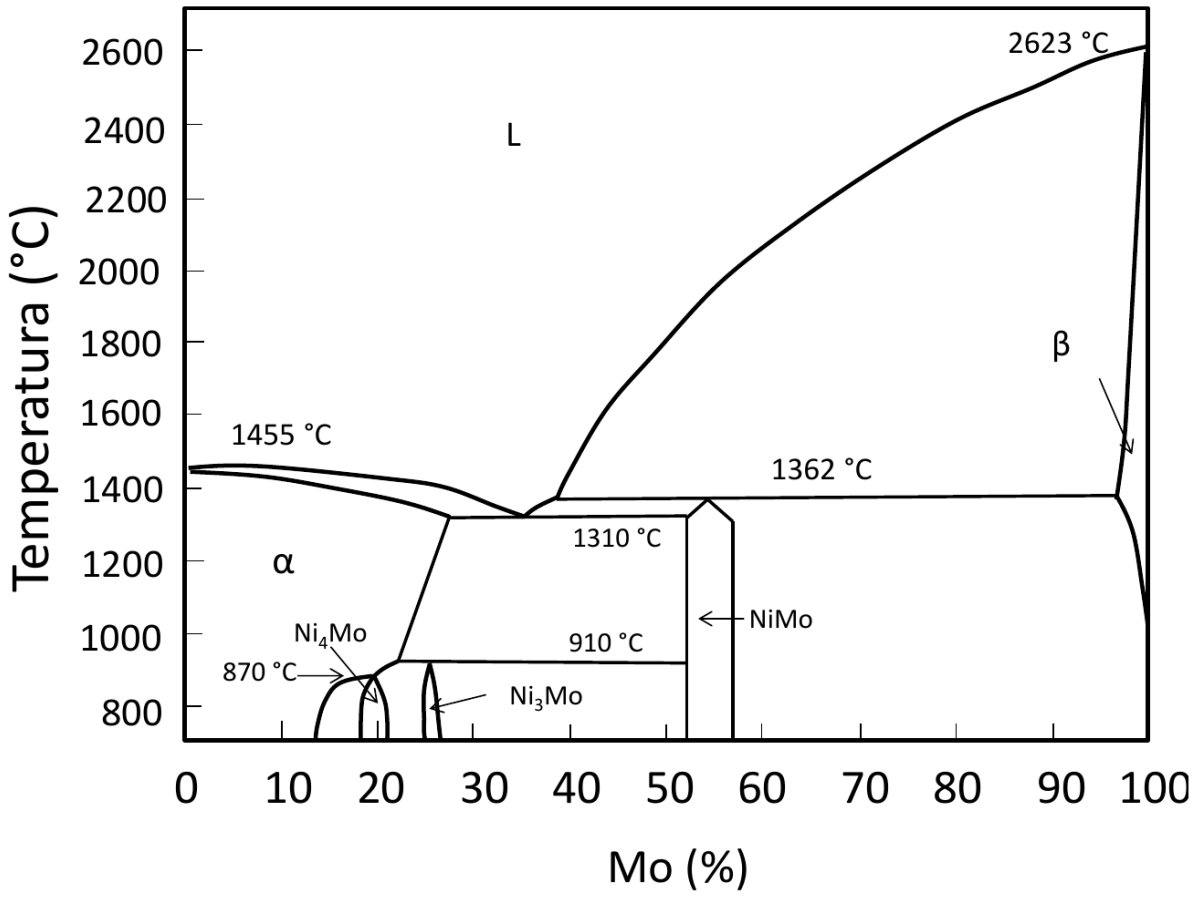


Figure 20. Ni-Mo phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 5).

Official Solution: a) eutectic reaction (1310 °C); peritectoid reaction (910 °C); peritectic reaction (1362 °C). b) 26% Mo. c) 4% Ni. d) 1455 °C; 2623 °C. e) C_{NiMo} **= 52%Mo;** C_{α} **= 27%Mo;** X_{α} **= 0.48;** X_{NiMo} **= 0.52. f)** $X_{NiMo,p}$ **= 0.25;** X_E **= 0.75**

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**, Ni-Mo diagram of the sheet (wt% Mo). Printed temperatures: 1455, 2623, 1362, 1310, 910 and 870 °C. Compositions are **readings of the figure** (±1 wt%).
2. α is the Ni-rich solid solution and β the Mo-rich one; NiMo is an intermediate phase with a narrow composition range; Ni3Mo and Ni4Mo are compounds that form in the solid.

Data read on the diagram

Line	Phases (wt% Mo)
1310 °C	α ≈ 27, L ≈ 36, NiMo ≈ 52
1362 °C	L ≈ 38, NiMo ≈ 55, β ≈ 96
910 °C	α ≈ 22, Ni_3Mo ≈ 25, NiMo ≈ 52

PHASE 2 · FRAMES & STRATEGY

Frame: the Ni-Mo diagram (wt% Mo, °C); one horizontal line per question, so no change of basis is needed. Reactions are named by their shape [Session 8 Slide 6]; amounts by the lever rule [Session 7 Slides 16, 32].

PHASE 3 · STEP-BY-STEP DERIVATION

a) Invariant reactions (on cooling)

T (°C)	Reaction	Type
1362	$L\,(38) + \beta\,(96) \rightarrow \mathrm{NiMo}\,(55)$	peritectic
1310	$L\,(36) \rightarrow \alpha\,(27) + \mathrm{NiMo}\,(52)$	eutectic
910	$\alpha\,(22) + \mathrm{NiMo}\,(52) \rightarrow \mathrm{Ni_3Mo}\,(25)$	peritectoid

At 870 °C the compound Ni4Mo appears inside the α field (the small dome); only α and Ni4Mo take part, so it is not a three-phase reaction.

b), c), d)

- Maximum solubility of **Mo in Ni**: about **27 wt% Mo** (the key writes 26 %), at the eutectic temperature, **1310 °C**.
- Maximum solubility of **Ni in Mo**: about **4 wt% Ni** (β at 96 wt% Mo), at the peritectic temperature, **1362 °C**.
- Melting points: **Ni 1455 °C**, **Mo 2623 °C**.

e) Phases of Ni-40 wt% Mo at 1309 °C Just below the eutectic: α (27) + NiMo (52), tie line length 25:

$$W_\alpha = \frac{C_{\text{NiMo}} - C_0}{C_{\text{NiMo}} - C_\alpha} = \frac{52 - 40}{52 - 27} = \frac{12}{25} = 0.48 \qquad W_{\text{NiMo}} = \frac{40 - 27}{25} = \frac{13}{25} = 0.52$$

f) Microconstituents at 1309 °C $36 < 40 < 52$: the alloy is **hypereutectic**, so primary NiMo forms first and the liquid of 36 wt% Mo becomes eutectic. Short tie line from 36 to 52 (length 16):

$$W_{\text{NiMo},p} = \frac{C_0 - C_e}{C_{\text{NiMo}} - C_e} = \frac{40 - 36}{52 - 36} = \frac{4}{16} = 0.25 \qquad W_E = \frac{52 - 40}{16} = \frac{12}{16} = 0.75$$

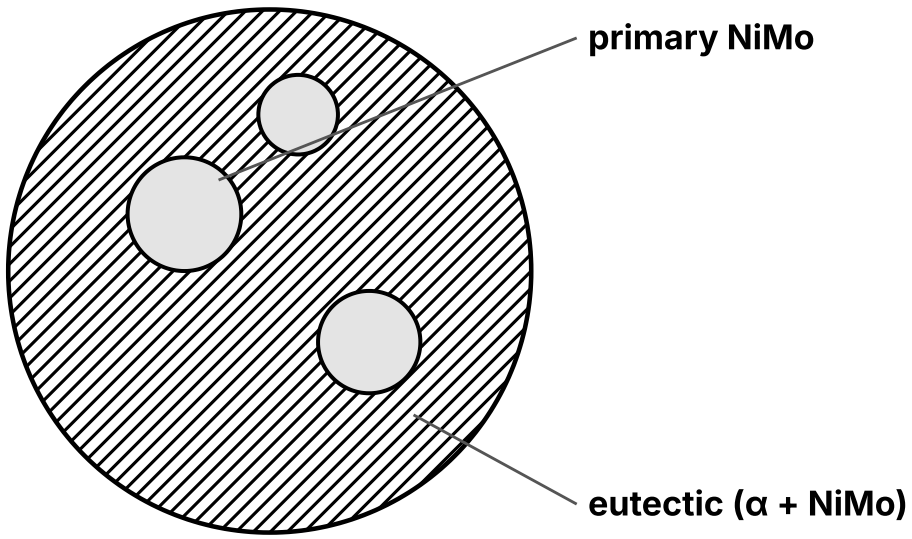


Figure 19. Ni-40 wt% Mo just below 1310 °C: primary NiMo (25 %) in the eutectic α + NiMo (75 %).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- Mass balance at 1309 °C: $0.48 \times 27 + 0.52 \times 52 = 12.96 + 27.04 = 40.0$ wt% Mo.
- Phases inside the eutectic: all the α (0.48) and NiMo $0.52 - 0.25 = 0.27$; $0.48 + 0.27 = 0.75 = W_E$. The eutectic alone is $(52 - 36)/(52 - 27) = 16/25 = 0.64$ α, and $0.75 \times 0.64 = 0.48$.

Interpretation The alloy is three quarters eutectic (α + NiMo) with a quarter of primary NiMo crystals. On further cooling, the α and NiMo of this alloy reach the 910 °C peritectoid, where they react to form Ni_3Mo.

Check against the official solution

All values agree with the key (0.48/0.52; 0.25/0.75; 1455 and 2623 °C; the three reactions). The key gives 26 % Mo for part (b) and uses 27 % Mo in part (e); the figure reads about 27 %.

Problem 13 | Problems T4_PhaseDiagrams I.pdf #13 | Ti-Si · congruent melting

Compounds and Cooling of a Ti-30 % Si Alloy

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 5

13. Given the following Ti-Si phase diagram answer the next questions: a) Indicate the phases or compounds with incongruent and congruent melting. b) Write down the invariant reactions, specifying their name, temperature, and composition of the phases present. c) For an alloy with 30% Si describe the solidification and cooling sequence to 600 °C. Describe specifically the microstructure that can be expected at 1329 °C, 1169 °C, and 600 °C. d) For the same alloy of part c) and the temperatures of 1329 °C and 1169 °C indicate phase composition, phase fractions and quantity of microconstituents present.

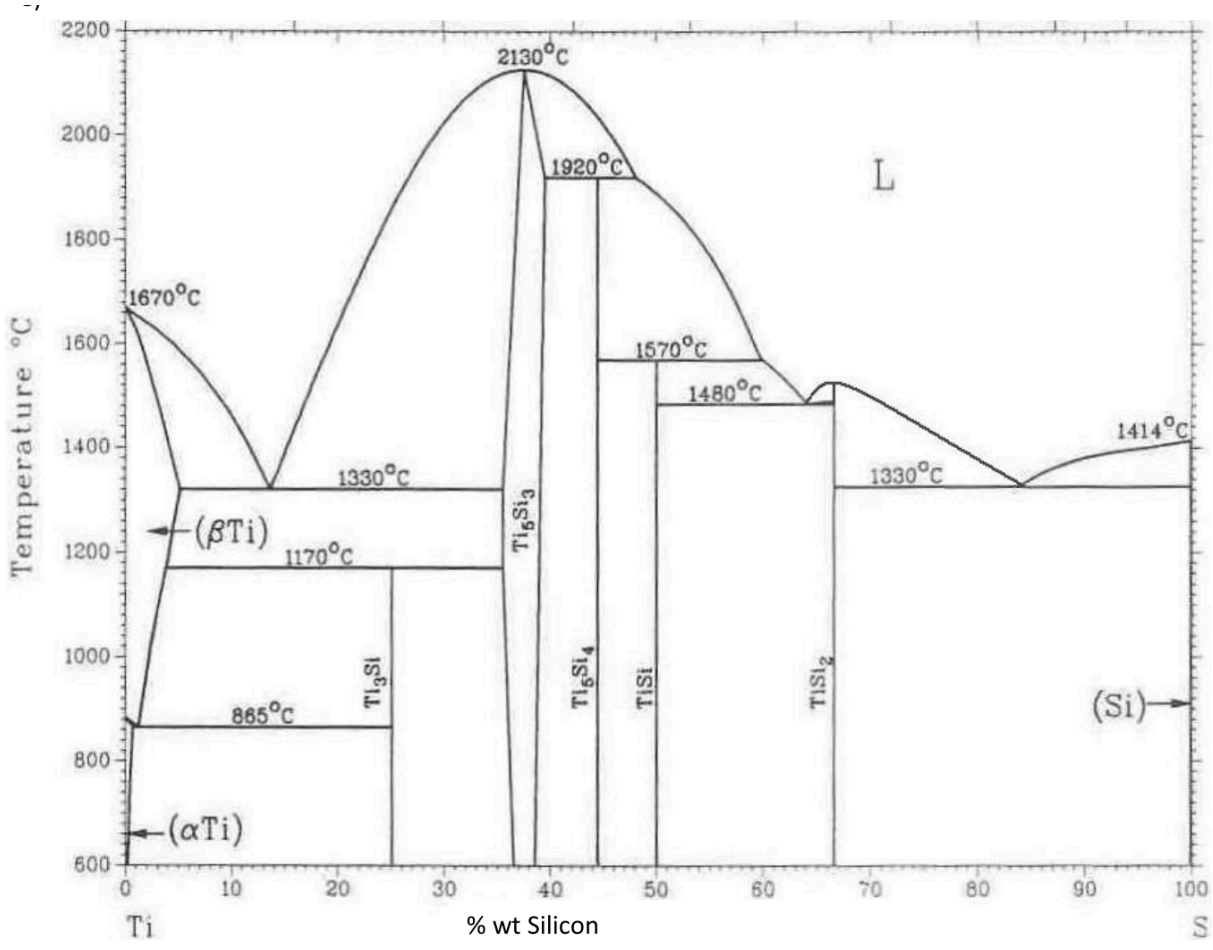


Figure 22. Ti-Si phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 6).

Official Solution: a) congruent melting: Ti_5Si_3 and $TiSi_2$; incongruent melting: Ti_3Si, Ti_5Si_4 and TiSi. b) Eutectic, T = 1330 °C; Eutectic, T = 1330 °C; Eutectic, T = 1480 °C; Peritectic, T = 1920 °C; Peritectic, T = 1570 °C; Peritectoid T = 1170 °C; Eutectoid, T = 865 °C. c) T = 1329 °C: $C_{\beta Ti}$ = 5.5% Si and $C_{Ti_5Si_3}$ = 35% Si, $x_{\beta Ti}$ = 0.17, $x_{Ti_5Si_3}$ = 0.83; T = 1169 °C: C_{Ti_3Si} = 25% Si and $C_{Ti_5Si_3}$ = 35% Si, $x_{Ti_5Si_3}$ = 0.5, x_{Ti_3Si} = 0.5. d) T = 1329 °C: $x_{eutectic}$ = 0.24, $x_{Ti_5Si_3,primary}$ = 0.76; T = 1169 °C phases and microconstituents are the same

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**, Ti-Si diagram of the sheet. Printed temperatures: 1670, 1414, 2130, 1920, 1570, 1480, 1330 (twice), 1170 and 865 °C. Compositions are **readings of the figure** (±1 %), on the axis as printed.
2. The axis is labelled "% wt Silicon", but the compounds sit at 25 (Ti_3Si), 37.5 (Ti_5Si_3), 44.4 (Ti_5Si_4), 50 (TiSi) and 66.7 ($TiSi_2$), their **atomic** percentages of Si; the numbers below are read on the axis as printed, as the key does.
3. Melting is congruent at a maximum of the liquidus and incongruent at a peritectic line [Session 8 Slide 17].

Data read on the diagram

Line	Phases (% Si)
1330 °C (Ti side)	βTi ≈ 5.5, L ≈ 14, Ti_5Si_3 ≈ 35
1170 °C	βTi ≈ 4, Ti_3Si 25, Ti_5Si_3 ≈ 35

PHASE 2 · FRAMES & STRATEGY

Frame: the Ti-Si diagram (% Si, °C); all the work is on horizontal lines and the vertical line of the alloy, so no change of basis is needed. Reactions are named by their shape [Session 8 Slide 6], amounts by the lever rule [Session 7 Slides 16, 32].

PHASE 3 · STEP-BY-STEP DERIVATION

a) Congruent and incongruent melting

- **Congruent:** Ti_5Si_3 (top of the large dome, 2130 °C) and $TiSi_2$ (small maximum near 1500 °C).
- **Incongruent:** Ti_5Si_4 and TiSi end at peritectic lines (1920 and 1570 °C), and Ti_3Si at the peritectoid line (1170 °C): on heating each decomposes into another phase plus liquid (or, for Ti_3Si, into two solids).

b) Invariant reactions (on cooling, % Si read on the axis)

T (°C)	Reaction	Type
1920	$L\,(\approx 48) + \mathrm{Ti_5Si_3}\,(\approx 38) \rightarrow \mathrm{Ti_5Si_4}\,(\approx 44)$	peritectic
1570	$L\,(\approx 59) + \mathrm{Ti_5Si_4}\,(\approx 44) \rightarrow \mathrm{TiSi}\,(50)$	peritectic
1480	$L\,(\approx 64) \rightarrow \mathrm{TiSi}\,(50) + \mathrm{TiSi_2}\,(\approx 67)$	eutectic
1330	$L\,(\approx 14) \rightarrow \beta\mathrm{Ti}\,(\approx 5.5) + \mathrm{Ti_5Si_3}\,(\approx 35)$	eutectic
1330	$L\,(\approx 84) \rightarrow \mathrm{TiSi_2}\,(\approx 67) + (\mathrm{Si})\,(\approx 100)$	eutectic
1170	$\beta\mathrm{Ti}\,(\approx 4) + \mathrm{Ti_5Si_3}\,(\approx 35) \rightarrow \mathrm{Ti_3Si}\,(25)$	peritectoid
865	$\beta\mathrm{Ti}\,(\approx 1) \rightarrow \alpha\mathrm{Ti}\,(\approx 0.5) + \mathrm{Ti_3Si}\,(25)$	eutectoid

c) Cooling of Ti-30 % Si to 600 °C

1. Liquid down to the liquidus of Ti5Si3 (roughly 1950 °C, a reading); there **primary Ti5Si3** starts to crystallise.
2. L + Ti5Si3 down to 1330 °C; the liquid becomes poorer in Si along the liquidus and reaches 14 % Si.
3. At 1330 °C the remaining liquid gives the **eutectic** βTi + Ti5Si3.
4. Between 1330 and 1170 °C: primary Ti5Si3 in a eutectic of βTi + Ti5Si3.
5. At 1170 °C the **peritectoid** βTi + Ti5Si3 → Ti3Si uses up all the βTi (the alloy, 30 %, lies between Ti3Si, 25 %, and Ti5Si3, 35 %), so below it the alloy is **Ti3Si + Ti5Si3**.
6. No other line is crossed down to 600 °C; the boundaries of both compounds are vertical, so nothing changes.

At 1329 °C (βTi 5.5 / Ti5Si3 35, length 29.5):

$$x_{\beta\mathrm{Ti}} = \frac{35 - 30}{35 - 5.5} = \frac{5}{29.5} = 0.17 \qquad x_{\mathrm{Ti_5Si_3}} = \frac{30 - 5.5}{29.5} = \frac{24.5}{29.5} = 0.83$$

At 1169 °C and at 600 °C (Ti3Si 25 / Ti5Si3 35, length 10):

$$x_{\mathrm{Ti_3Si}} = \frac{35 - 30}{35 - 25} = \frac{5}{10} = 0.5 \qquad x_{\mathrm{Ti_5Si_3}} = \frac{30 - 25}{10} = 0.5$$

d) Microconstituents At 1329 °C: primary Ti5Si3 and eutectic. Short tie line from the eutectic liquid (14) to Ti5Si3 (35), length 21:

$$x_{\mathrm{Ti_5Si_3},p} = \frac{30 - 14}{35 - 14} = \frac{16}{21} = 0.76 \qquad x_E = \frac{35 - 30}{21} = \frac{5}{21} = 0.24$$

At 1169 °C: the peritectoid has turned the βTi (which was inside the eutectic) and part of the Ti_5Si_3 into Ti_3Si; the structure is Ti_3Si and Ti_5Si_3, in the same amounts as the phases, 0.5 and 0.5.

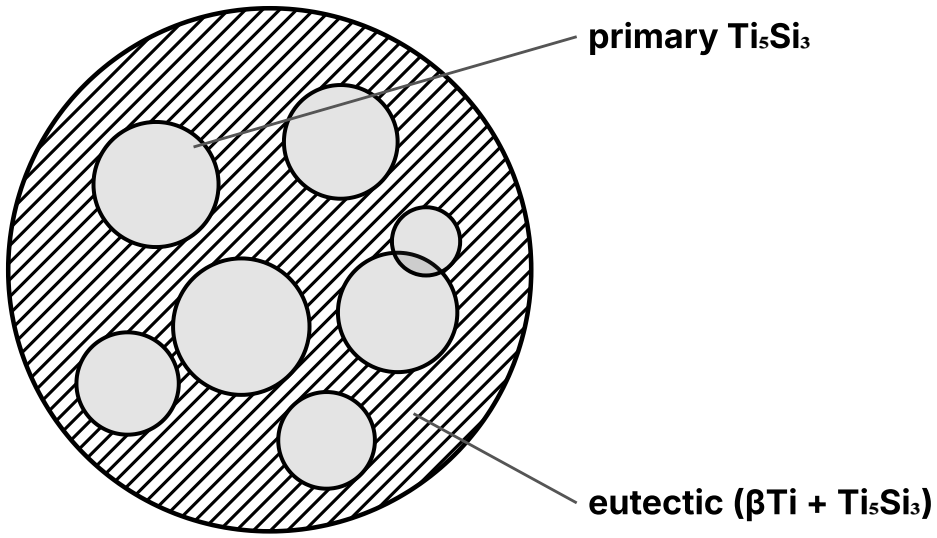


Figure 21. Ti-30 wt% Si just below 1330 °C: primary Ti_5Si_3 (76 %) in the eutectic βTi + Ti_5Si_3 (24 %).

PHASE 4 · CHECKS & INTERPRETATION

Checks

- 1329 °C: $0.17 \times 5.5 + 0.83 \times 35 = 0.94 + 29.06 = 30.0$ % Si.
- 1169 °C: $0.5 \times 25 + 0.5 \times 35 = 30$ % Si.
- Eutectic at 1329 °C: it holds all the βTi (0.17) and Ti_5Si_3 $0.83 - 0.76 = 0.07$; $0.17 + 0.07 = 0.24 = x_E$.

Interpretation The Ti-rich side combines a eutectic, a peritectoid and a eutectoid. The 30 % alloy ends up made only of two silicides, which are hard and brittle compounds.

Check against the official solution

All values agree with the key (0.17/0.83, 0.5/0.5, 0.24/0.76; the classification of the compounds and the seven reactions).

Problem 14 | Problems T4_PhaseDiagrams I.pdf #14 | Invariant reactions

Reading a Hypothetical Diagram with Several Compounds

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 6

14. In the following graph, a hypothetical phase diagram is represented, identify and explain the following: a) The existing phases in each phase field of the diagram. b) The invariant reactions writing the temperature and the reaction's equation, the composition of each phase inside the reaction and its name. c) The congruent and incongruent fusion phases.

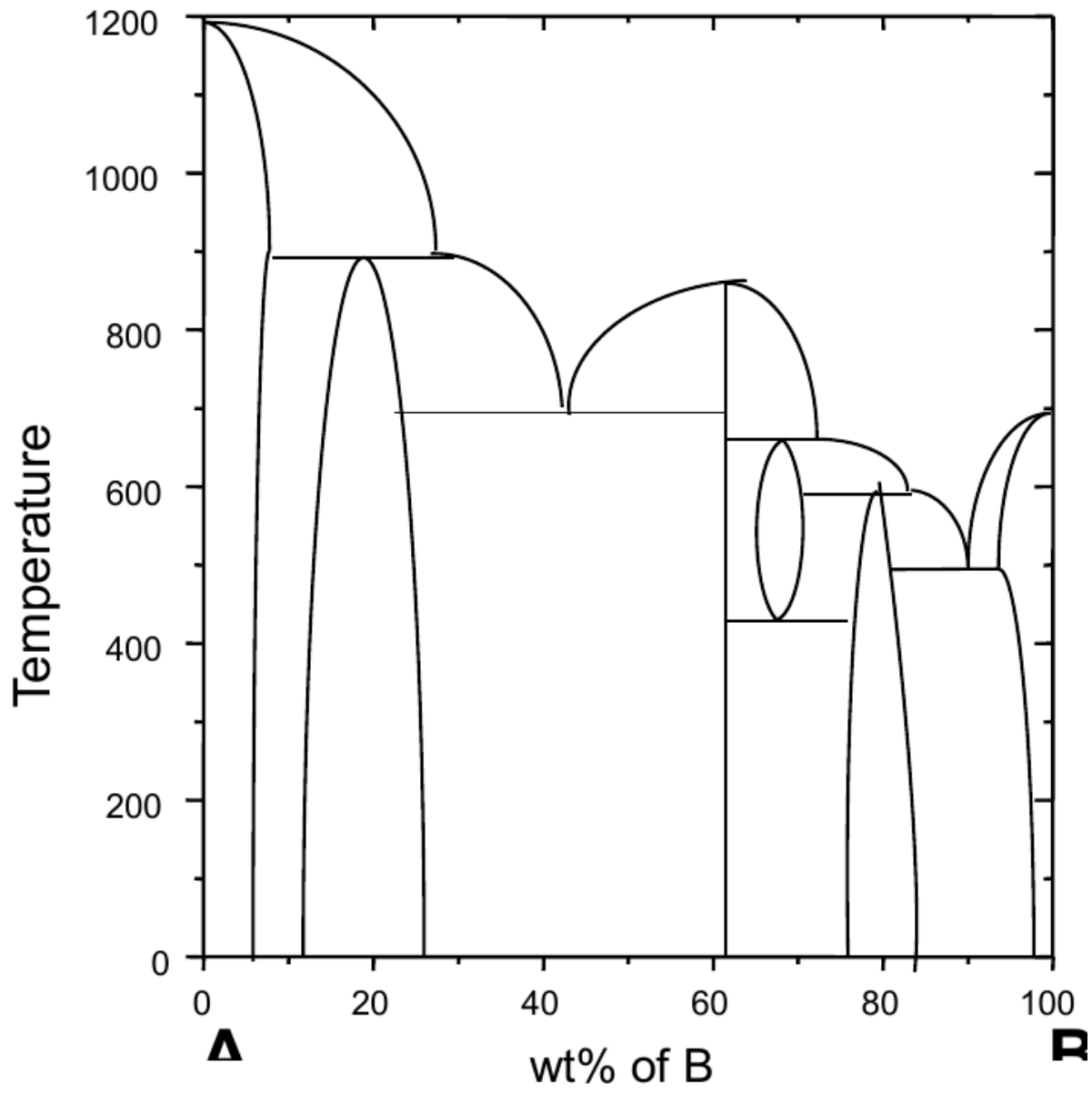


Figure 23. Hypothetical A-B phase diagram (Problems T4_PhaseDiagrams I.pdf, p. 6).

Official Solution: β incongruent melting; γ congruent melting; δ incongruent melting; ε incongruent melting

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**; the diagram has no printed values except the axes (wt% B, temperature in °C), so every number below is a **reading of the figure**, to about ±1 wt% and ±10 °C.
2. Phases are named as in the official solution (p. 37): terminal solid solutions α (A side) and ζ (B side), intermediate phases β, γ (a vertical line: a compound of fixed composition at about 62 % B), δ and ε, and the liquid L.
3. Each horizontal line is an invariant reaction, named by its shape [Session 8 Slide 6]; melting is congruent at a maximum of the liquidus and incongruent at a peritectic line [Session 8 Slide 17].

PHASE 2 · FRAMES & STRATEGY

Frame: composition (wt% B) against temperature. The diagram is read on horizontal lines, from the top down and from left to right; no change of basis is involved.

Method. (a) Single-phase fields are the regions bounded by curved lines that touch the axis or a dome; each two-phase field takes the phases on its two sides along a horizontal tie line [Session 7 Slide 15]. (b) For each horizontal line: the three phases at its ends and middle point, and the shape. (c) Where each intermediate phase ends at the top.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phase fields

Field	Phases
top	L
A side, below the liquidus	α + L, then α
around 20 % B	β; α + β on its left; β + L above the 700 °C line on its right
23-62 % B, below 700 °C	β + γ
43-62 % B, above 700 °C	γ + L
62 % B	γ (vertical line)
62-72 % B, above 660 °C	γ + L
closed field near 68 % B, 430-660 °C	δ, with γ + δ, δ + L and δ + ε around it
62-80 % B, below 430 °C	γ + ε
around 80 % B	ε, with ε + L above it

Field	Phases
81-94 % B, below 495 °C	ε + ζ
B side	ζ + L, then ζ

b) Invariant reactions (on cooling, wt% B)

T (°C)	Reaction	Name
≈ 895	$\alpha\,(8) + L\,(28) \rightarrow \beta\,(19)$	peritectic
≈ 700	$L\,(43) \rightarrow \beta\,(23) + \gamma\,(62)$	eutectic
≈ 660	$\gamma\,(62) + L\,(72) \rightarrow \delta\,(68)$	peritectic
≈ 595	$\delta\,(71) + L\,(84) \rightarrow \varepsilon\,(80)$	peritectic
≈ 495	$L\,(90) \rightarrow \varepsilon\,(81) + \zeta\,(94)$	eutectic
≈ 430	$\delta\,(68) \rightarrow \gamma\,(62) + \varepsilon\,(76)$	eutectoid

c) Melting

- **γ melts congruently** at about 865 °C: it is at the maximum of its liquidus dome, and the liquid formed has its composition.
- **β, δ and ε melt incongruently**: each ends at a peritectic line (895, 660 and 595 °C), where on heating it gives a liquid plus a different solid (β → α + L, δ → γ + L, ε → δ + L).
- The pure components A (≈ 1200 °C) and B (≈ 695 °C) also melt congruently, like any pure metal [Session 8 Slide 17].

PHASE 4 · CHECKS & INTERPRETATION

Checks

- On every eutectic and eutectoid line the parent phase (L or δ) lies **between** the two products; on every peritectic line the product lies between the two reactants, with the liquid at one end [Session 8 Slide 6].
- Phase rule: three phases on each horizontal line, $F = 2 - 3 + 1 = 0$ [Session 7 Slide 13].
- δ is a closed field: it is created by a peritectic at 660 °C and destroyed by a eutectoid at 430 °C, so it exists only between those temperatures.

Interpretation The diagram is a chain of simple diagrams joined at the compounds. A phase that touches the liquid only at a horizontal line cannot exist as a liquid of its own composition, which is the meaning of incongruent melting.

Check against the official solution

Part (c) agrees with the key (β, δ, ε incongruent; γ congruent). The key gives no temperatures or compositions for part (b); they are readings of the figure here. The field labels agree with the annotated diagram of Solution problems Phase diagrams I.pdf, p. 37.

Problem 15 | Problems T4_PhaseDiagrams I.pdf #15 | Ti-B · peritectics

The Ti-B System and Ti-10 % B and Ti-18 % B Alloys

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 7

15. Given the phase diagram of the figure, answer the following questions: a. Write down the invariant reactions, indicating the name, temperature and composition of the phases involved. b. List the congruent and incongruent melting phases or compounds. c. For an alloy with 10% B, indicate the composition of the phases, the amount of phases and the amount of microconstituents at 1000 °C and 500 °C. d. Describe the solidification sequence of an alloy with 18% B at a very slow cooling from the liquid state up to 500 °C.

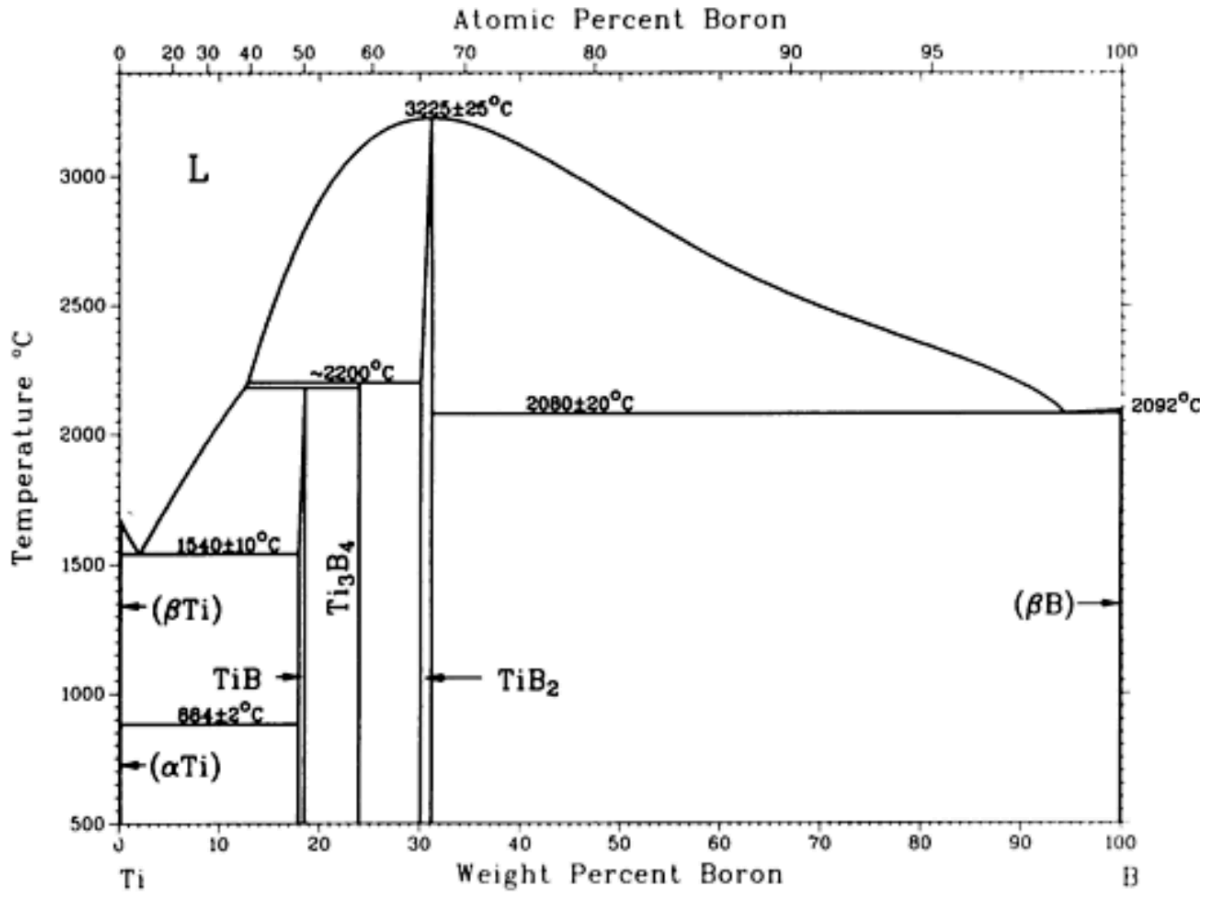


Figure 25. Ti-B phase diagram in weight percent (Problems T4_PhaseDiagrams I.pdf, p. 7).

Official Solution: a) peritectic (2200 °C); peritectic (2190 °C); eutectic (1540 °C); eutectic (2080 °C). b) Ti_3B_4 → L + TiB_2. c) 1000 °C: $C_{(\beta Ti)}$ = 0%B; C_{TiB} = 18%B; $X_{(\beta Ti)}$ = 0.44; $X_{(TiB)}$ = 0.56; eutectic (x = 50%) and primary TiB (x = 50%). 500 °C: $C_{(\alpha Ti)}$ = 0%B; C_{TiB} = 18%B; $X_{(\alpha Ti)}$ = 0.44; $X_{(TiB)}$ = 0.56; eutectic (x = 50%) and primary TiB (x = 50%)

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**, Ti-B diagram of the sheet, read on its **weight percent** scale (bottom axis). Printed values: 1540, 884, ~2200, 2080, 3225 and 2092 °C. Compositions are **readings of the figure** (±1 wt%).
2. Boron is practically insoluble in Ti: (βTi) and (αTi) are taken as 0 wt% B, as the key does.
3. TiB, Ti_3B_4 and TiB_2 are compounds (Slide 15 of Session 8 shows the same system on the atomic scale).

Data read on the diagram

Line	Phases (wt% B)
1540 °C eutectic	(βTi) ≈ 0, L ≈ 2, TiB ≈ 18
~2200 °C	L ≈ 13, Ti_3B_4 ≈ 24, TiB_2 ≈ 31
~2190 °C	L ≈ 13, TiB ≈ 18, Ti_3B_4 ≈ 24
2080 °C eutectic	TiB_2 ≈ 31, L ≈ 94, (βB) ≈ 100

PHASE 2 · FRAMES & STRATEGY

Frame: the Ti-B diagram (wt% B, °C); horizontal tie lines and the vertical line of each alloy; no change of basis. Names of reactions from their shape [Session 8 Slide 6]; amounts from the lever rule [Session 7 Slides 16, 32].

PHASE 3 · STEP-BY-STEP DERIVATION

a) Invariant reactions (on cooling)

T (°C)	Reaction	Name
~2200	$L\,(13) + \mathrm{TiB_2}\,(31) \rightarrow \mathrm{Ti_3B_4}\,(24)$	peritectic
~2190	$L\,(13) + \mathrm{Ti_3B_4}\,(24) \rightarrow \mathrm{TiB}\,(18)$	peritectic
2080	$L\,(94) \rightarrow \mathrm{TiB_2}\,(31) + (\beta\mathrm{B})\,(100)$	eutectic
1540	$L\,(2) \rightarrow (\beta\mathrm{Ti})\,(0) + \mathrm{TiB}\,(18)$	eutectic

At 884 °C the titanium changes from (βTi) to (αTi) (its allotropic change, Topic 2); TiB does not take part.

b) Melting

- **TiB_2 melts congruently**, at the top of the liquidus, 3225 °C.

- **Ti_3B_4 melts incongruently** ($Ti_3B_4 \rightarrow L + TiB_2$ on heating at ~2200 °C), and so does **TiB** ($TiB \rightarrow L + Ti_3B_4$ at ~2190 °C): both end at peritectic lines [Session 8 Slide 17].

c) Ti-10 wt% B at 1000 °C and 500 °C Phases at 1000 °C: (βTi) 0 and TiB 18 (length 18):

$$X_{(\beta\mathrm{Ti})} = \frac{18-10}{18-0} = \frac{8}{18} = 0.44 \qquad X_{\mathrm{TiB}} = \frac{10-0}{18} = \frac{10}{18} = 0.56$$

At 500 °C the same, with (αTi) instead of (βTi): 0.44 and 0.56.

Microconstituents. $2 < 10 < 18$: hypereutectic, primary TiB and eutectic. Short tie line from the eutectic liquid (2) to TiB (18), length 16:

$$X_{\mathrm{TiB},p} = \frac{10-2}{18-2} = \frac{8}{16} = 0.50 \qquad X_E = \frac{18-10}{16} = 0.50$$

The same at 1000 and 500 °C (the 884 °C change affects only the Ti inside the eutectic).

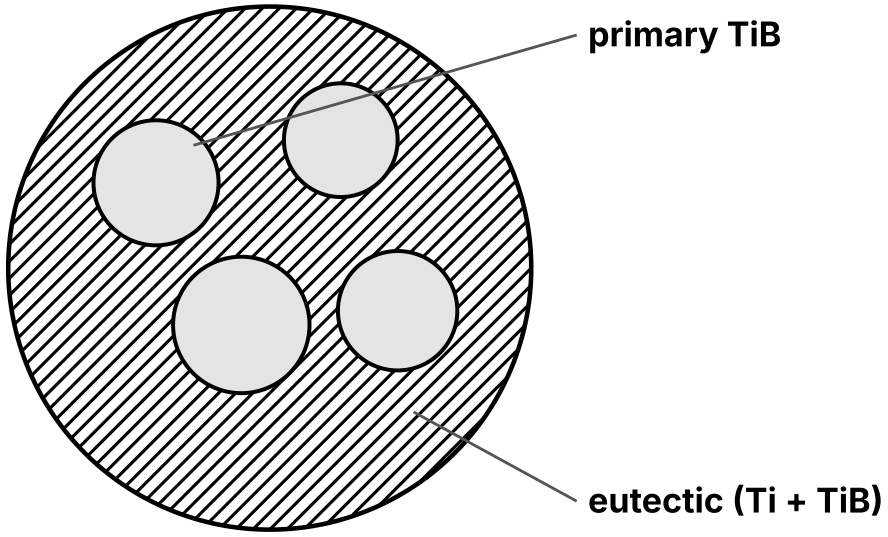


Figure 24. Ti-10 wt% B below 1540 °C: primary TiB (50 %) in the eutectic Ti + TiB (50 %).

d) Ti-18 wt% B, very slow cooling to 500 °C

1. **Liquid**, down to the liquidus on the TiB_2 side of the dome (roughly 2500 °C at 18 wt% B, a reading).
2. **L + TiB_2**: primary TiB_2 crystallises; the liquid becomes poorer in B, down to 13 wt% at ~2200 °C.
3. **~2200 °C, peritectic** $L + TiB_2 \rightarrow Ti_3B_4$: since $13 < 18 < 24$, all the TiB_2 is used up and the alloy is L + Ti_3B_4.
4. **~2190 °C, peritectic** $L + Ti_3B_4 \rightarrow TiB$: the alloy has exactly the composition of TiB, so liquid and Ti_3B_4 are both used up: **100 % TiB**.

5. From ~2190 to 500 °C nothing changes: the alloy stays single-phase TiB.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- (c): $0.44 \times 0 + 0.56 \times 18 = 10$ wt% B; the eutectic (0.50) contains all the Ti (0.44) and $0.56 - 0.50 = 0.06$ of TiB, $0.44 + 0.06 = 0.50$.
- (d): in equilibrium each peritectic is complete; with faster cooling the TiB would surround the Ti_3B_4 (and that the TiB_2) and the reactions would stay incomplete [Session 8 Slide 10].

Interpretation A Ti-B alloy with 10 % B is half eutectic (Ti + TiB) and half primary TiB, a hard boride; the 18 % B alloy is a single compound, reached through two peritectic reactions in a row.

Check against the official solution

Parts (a), (b) and (c) agree with the key (0.44/0.56 and 50 %/50 % at both temperatures). Part (b) of the key lists only Ti_3B_4; TiB is also incongruent and TiB_2 congruent, as shown above. The key gives no answer for part (d).